

16/11/8

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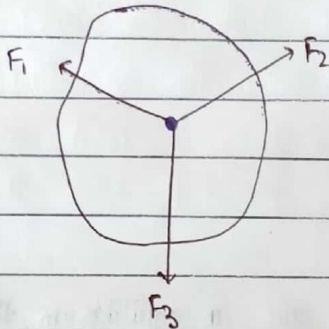
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UNIT-1

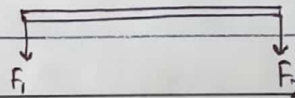
Force System

Resultant of several forces can be determined by:

- Graphical method
- Analytical method



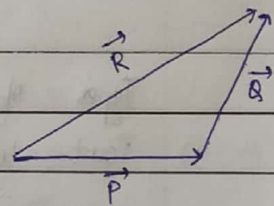
Concurrent forces
 $\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$



Parallel forces
 $\vec{R} = \vec{F}_1 + \vec{F}_2$

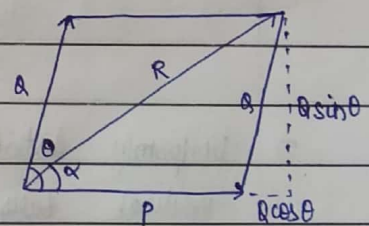
Triangle law:

$$\vec{R} = \vec{P} + \vec{Q}$$

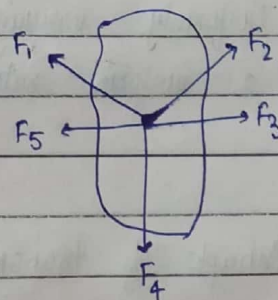
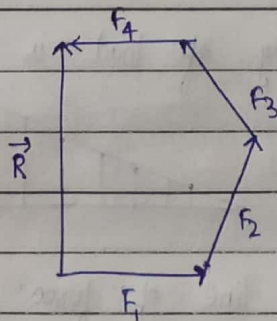


Parallelogram law:

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$



Polygon law:

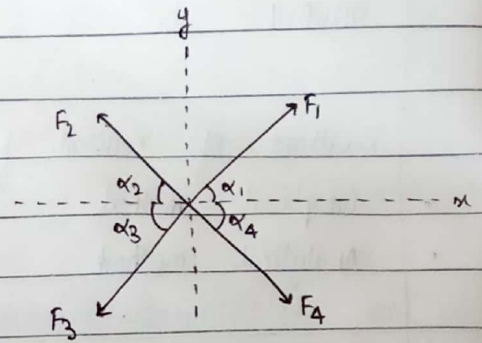
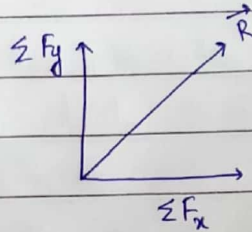


$\vec{R}$ is equal and opposite to F_5 .

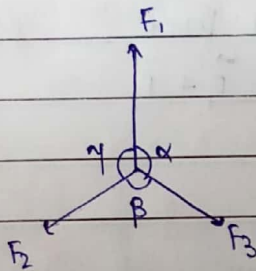
$$\sum F_x = F_1 \cos \alpha_1 + F_4 \cos \alpha_4 - F_2 \cos \alpha_2 - F_3 \cos \alpha_3$$

$$\sum F_y = F_1 \sin \alpha_1 + F_2 \sin \alpha_2 - F_3 \sin \alpha_3 - F_4 \sin \alpha_4$$

$$|\vec{R}| = \sqrt{F_x^2 + F_y^2}$$



Lamme's Theorem:

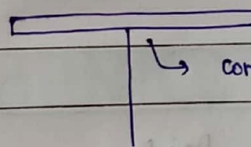


If F_1 , F_2 and F_3 are in equilibrium, then

$$\frac{F_1}{\sin \beta} = \frac{F_2}{\sin \alpha} = \frac{F_3}{\sin \gamma}$$

Types of supports:

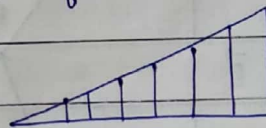
1) Point load:



concentrated or point load

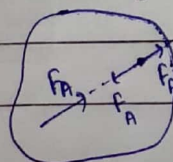
2) Uniformly distributed load: eg. the beam on ceiling is supported on vertical bars

3) Uniformly varying load: Density of load is increasing / decreasing at a uniform rate.



Principle of transmissibility of line of force:

No change is observed.

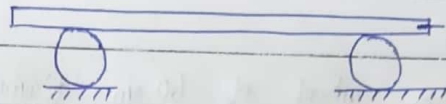


Types of supports :

- 1) Frictionless
- 2) Roller
- 3) Hinged
- 4) Fixed

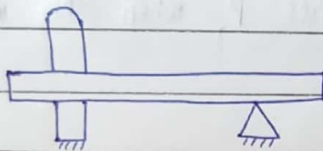


Frictionless Support



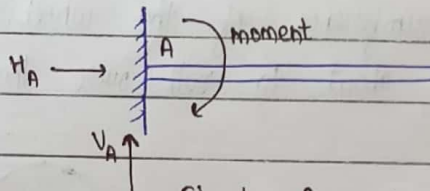
Roller Support

It only stops vertical movement not horizontal. As a result, the length of load free.



Hinged Support

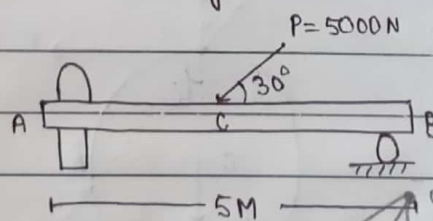
- It checks both horizontal and vertical movement.
- Permits movement about axis.



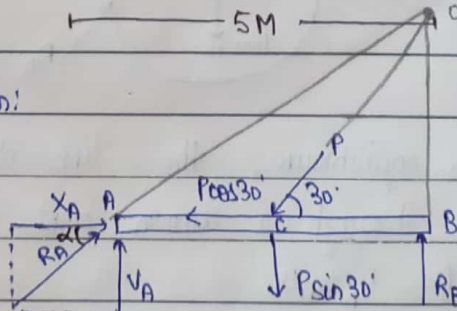
Fixed Support

Moment counteracts the moment produced by a weight present on the beam.

- Q. A force of 5000 N is applied at the centre C of the beam of length 5 m. Find the reactions at the hinge and roller support, if the beam is in eq.



Solⁿ Free Body Diagram:



Since the body is in equilibrium, all the forces should meet at one point.

$$R_A = \sqrt{X_A^2 + V_A^2}$$

$$\sum M_A = 0$$

$$P \sin 30^\circ \times 2.5 = R_B \times 5$$

clockwise anti-clockwise

$$\Rightarrow R_B = 1250 \text{ N}$$

$$\sum F_x = 0$$

$$X_A = P \cos 30^\circ$$

$$\Rightarrow X_A = 2500\sqrt{3} \text{ N}$$

$$\sum F_y = 0$$

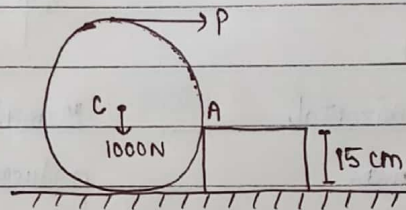
$$V_A + R_B = P \sin 30^\circ$$

$$\Rightarrow V_A = 1250 \text{ N}$$

$$\tan \alpha = \frac{V_A}{X_A} = \frac{1250}{2500\sqrt{3}} = \frac{1}{2\sqrt{3}}$$

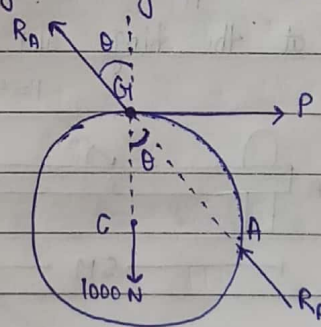
$$\alpha = \tan^{-1}\left(\frac{1}{2\sqrt{3}}\right)$$

8. A uniform wheel of 60 cm diameter and weighing 1000 N, rests against a rectangular block 15 cm high and lying on a horizontal plane. It is to be pulled over this block by a horizontal force P , applied to the end of a string wrapped around the circumference of the wheel. Find the force P , when the wheel is just about to roll over the block.



Sol.ⁿ When the wheel is just about to roll over the block, the normal force applied by the ground on the wheel becomes zero.

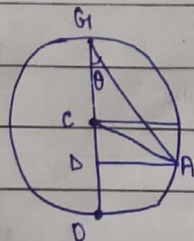
Free Body diagram:



Since body is in equilibrium, the line of action of all 3 forces should pass through a common point.

$$\sum F_x = 0 \Rightarrow P = R_A \sin \theta$$

$$\sum F_y = 0 \Rightarrow R_A \cos \theta = 1000$$



$$CB = 15 \text{ cm} \quad [\because OD = 15 \text{ cm}]$$

$$AC = 30 \text{ cm}$$

$$AB = \sqrt{30^2 - 15^2}$$

$$GD = 45 \text{ cm}$$

$$\tan \theta = \frac{AD}{GD} = \frac{\sqrt{675}}{45} = \frac{15\sqrt{3}}{45} = \frac{1}{\sqrt{3}}$$

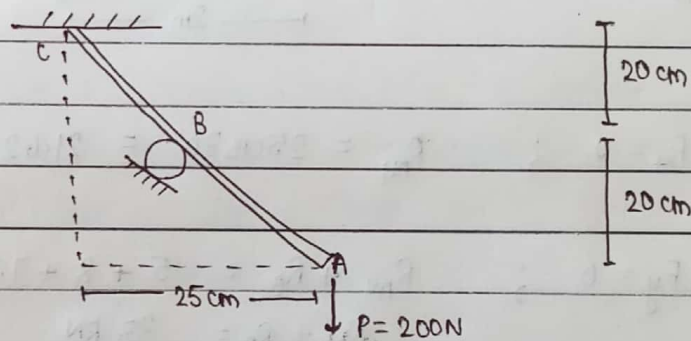
$$\theta = 30^\circ$$

$$\text{So, } R_A \cos 30^\circ = 1000$$

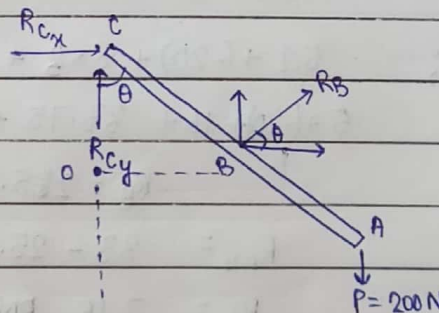
$$R_A \cdot \frac{\sqrt{3}}{2} = 1000 \Rightarrow R_A = \frac{2000}{\sqrt{3}} = 1156 \text{ N}$$

$$P = R_A \sin 30^\circ = 578 \text{ N}$$

- Q. A force of $P = 200 \text{ N}$ is applied at the end of a rod ABC which is supported by a hinge at C and rests against a roller support of B. Find the reactions at C and B.



Sol.ⁿ Free Body diagram :



$$\tan \theta = \frac{25}{40} = \frac{5}{8}$$

$$\sum M_C = 0$$

$$R_B \cdot BC = 200 \cdot 25$$

$$R_B = 212.13 \text{ N}$$

$$\sum F_x = 0 ;$$

$$\sum F_y = 0 ;$$

$$R_{Cx} + R_B \cos \theta = 0$$

$$R_{Cx} = -212.13 \times \cos 32^\circ = -180.8 \text{ N}$$

$$R_{Cy} + R_B \sin \theta = 200$$

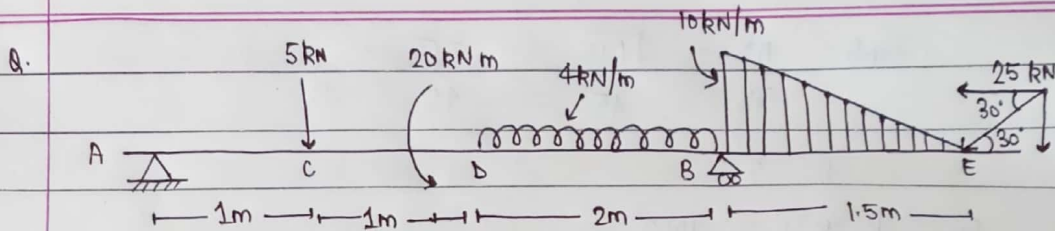
$$R_{Cy} = 200 - 112.42 = 88 \text{ N}$$

$$OB = 12.5 \text{ cm}$$

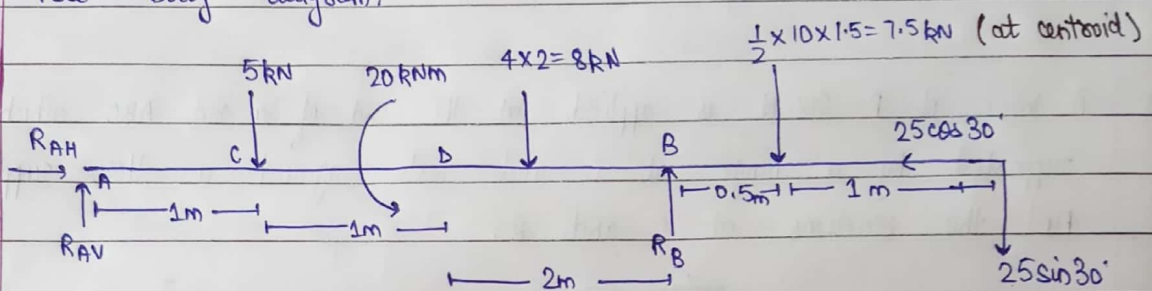
$$BC = \sqrt{OB^2 + OC^2}$$

$$BC = \sqrt{12.5^2 + 20^2}$$

$$BC = 23.57 \text{ cm}$$



Sol.ⁿ 4 kN/m , $10\text{ kN/m} \rightarrow$ intensity
Free Body diagram;



$$\sum F_x = 0 ; \quad R_{AH} = 25 \cos 30^\circ = 21.625 \text{ kN}$$

$$\sum F_y = 0 ; \quad R_{AV} + R_B = 5 + 8 + 7.5 + 25 \sin 30^\circ$$

$$R_{AV} + R_B = 33 \text{ kN}$$

$$\sum M_A = 0 ; \quad 5 \cdot 1 + (-20) + 8 \times 2 + 7.5 \times 4.5 + 25 \sin 30^\circ \times 5.5 = 4R_B$$

$$5 + (-20) + 16 + 33.75 + 68.75 = 4R_B$$

$\uparrow$ upward $\rightarrow$ anti-clockwise

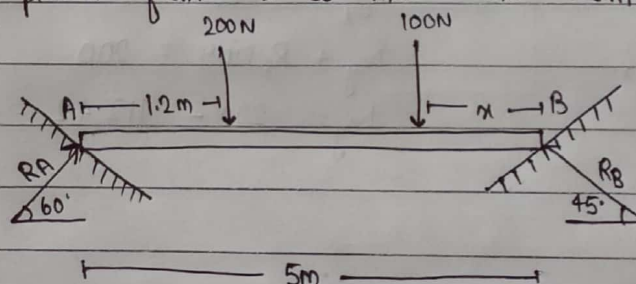
$$R_B = 25.85 \text{ kN}$$

$\downarrow$ downward $\rightarrow$ clockwise

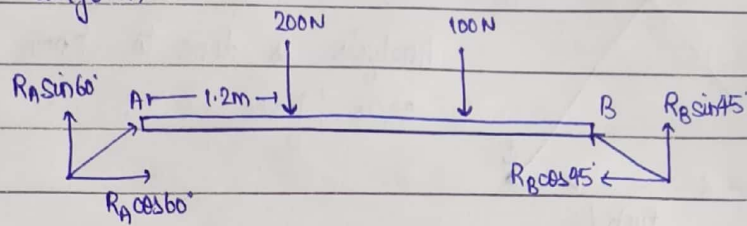
$$R_{AV} = 33 - 25.85$$

$$R_{AV} = 7.15 \text{ kN}$$

B. A weightless bar AB is placed in horizontal position on smooth inclined planes. Calculate the distance x at which the 100 N load should be placed from B so that the beam remains in equilibrium.



Solⁿ Free Body diagram:



$$\sum F_x = 0 ; \quad R_A \cos 60^\circ = R_B \cos 45^\circ$$

$$R_A = 1.414 R_B$$

$$\sum F_y = 0 ; \quad R_A \sin 60^\circ + R_B \sin 45^\circ = 300$$

$$\frac{\sqrt{3} \cdot (1.414) R_B}{2} + \frac{R_B}{\sqrt{2}} = 300$$

$$R_B = 155.36 \text{ N}$$

$$R_A = 219.68 \text{ N}$$

$$\sum M_B = 0 ; \quad 100x + 200 \times [5 - (1.2 + x)] = R_A \cdot 5 \sin 60^\circ$$

$$100x + 200 \times 3.8 = 219.68 \sin 60^\circ \times 5$$

$$100x + 760 = 950$$

$$100x = 190$$

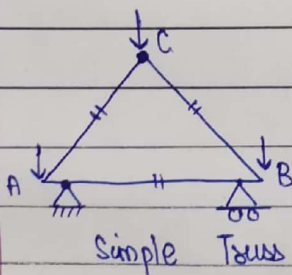
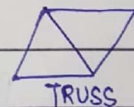
$$x = 1.9 \text{ m}$$

Analysis of a Truss

Truss is an engineering structure made of various bars having uniform cross-section. It is used for supporting the load.

1) Roof Truss

2) Bridge Truss

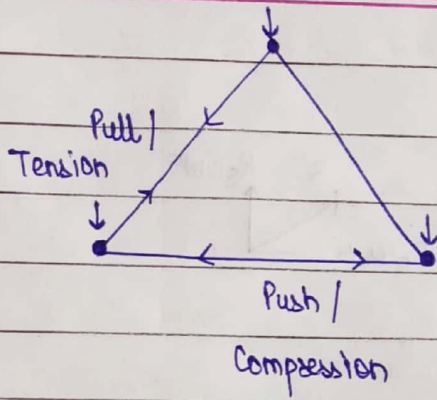


Simple Truss

We always provide a hinge and a roller support. Both supports can't be hinge or roller.

In a truss, load is only applied at the joints; not in the member.
 member of truss

- If a member is in equilibrium under 2 forces only, it is called 2 force body.



Analysis is done to know the internal stress / force.

- Perfect truss is the one, which is statically determinate. It is in static equilibrium i.e. $\sum F_x = 0$, $\sum F_y = 0$, $\sum M_A = 0$.
- There are 2 types of truss on the basis of rigidity:
 - (i) Over rigid - contains more members than what is required
 - (ii) Under rigid - contains less members than required

Extra members in over rigid truss do not provide rigidity, rather they add to cost and are useless.

- For a perfect truss,

$$m = 2j - 3$$

$m \rightarrow$ no. of members

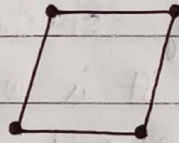
$j \rightarrow$ no. of joints



Perfect truss

$$m = 3$$

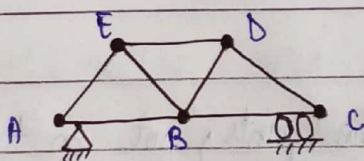
$$2j - 3 = 3$$



Under rigid truss

$$m = 4$$

$$2j - 3 = 5$$

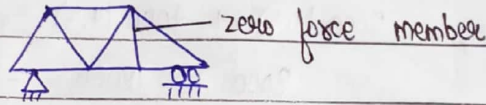


Perfect Truss

$$m = 7$$

$$2j - 3 = 7$$

- Sometimes, a member of a truss does not take any load, which means its internal force would also be zero. It is called zero-force member.



* Assumptions for analysis of Truss:

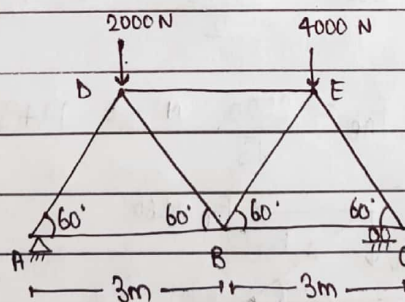
- 1) Joints are frictionless.
- 2) Load is applied only at the joints.
- 3) Each member of a truss is a 2 force body.
- 4) Resultant of all forces should be zero.
- 5) Weight of the member is neglected.
- 6) All truss are statically determinable.

* Methods to find internal stress:

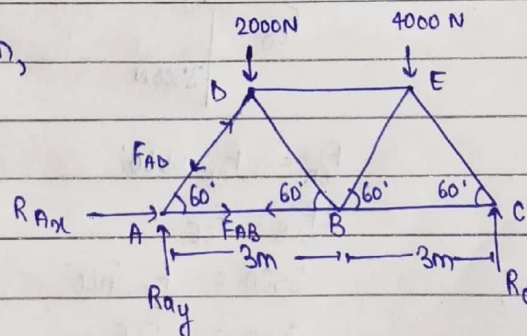
- 1) Method of joints
- 2) Method of sections
- 3) Graphical method

1. Method of joints

Q. Find the internal forces.



Sol.ⁿ Free body diagram,



$$\sum F_x = 0 ; R_{Ax} = 0$$

$$\sum F_y = 0 \quad ; \quad R_{Ay} + R_c = 6000$$

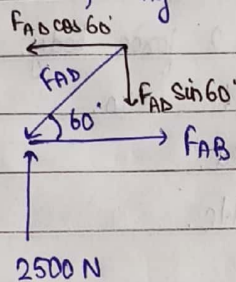
$$\sum M_A = 0 \quad ; \quad 2000(1.5) + 4000(4.5) = R_c \times 6$$

$$3000 + 18000 = 6R_c$$

$$6R_c = 21000$$

$$R_c = \frac{21000}{6} = 3500 \text{ N}$$

So, $R_{Ay} = 2500 \text{ N}$
at joint A,



Always start with a joint, which has only 2 unknowns.

$$\sum F_x = 0 \quad ; \quad F_{AB} = F_{AD} \cos 60^\circ$$

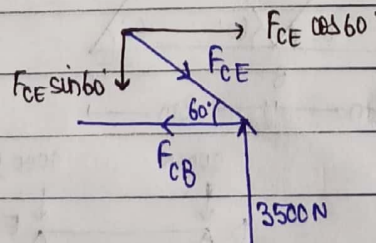
$$F_{AB} = \frac{F_{AD}}{2} \quad \text{--- (1)}$$

$$\sum F_y = 0 \quad ; \quad 2500 = F_{AD} \sin 60^\circ$$

$$2500 = \frac{\sqrt{3} F_{AD}}{2} \Rightarrow F_{AD} = \frac{5000}{\sqrt{3}} \text{ N} = 2887 \text{ N (compression)}$$

$$F_{AB} = \frac{2500}{\sqrt{3}} \text{ N} = 1443 \text{ N (tension)}$$

at joint C,



$$\sum F_x = 0 \quad ; \quad F_{CB} = F_{CE} \cos 60^\circ$$

$$2F_{CB} = F_{CE}$$

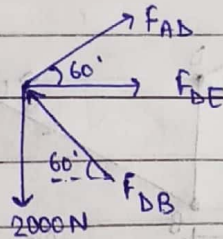
$$\sum F_y = 0 \quad ; \quad 3500 = F_{CE} \sin 60^\circ$$

$$7000 = \sqrt{3} F_{CE}$$

$$F_{CE} = 4041 \text{ N (compression)}$$

$$F_{CB} = 2020.5 \text{ N (tension)}$$

at joint D,



$$\sum F_x = 0;$$

$$F_{AD} \cos 60^\circ + F_{DE} = F_{DB} \cos 60^\circ$$

$$F_{AD} + 2F_{DE} = F_{DB}$$

$$2887 + 2F_{DE} = F_{DB}$$

$$\sum F_y = 0;$$

$$2000 = F_{DB} \sin 60^\circ + F_{AD} \sin 60^\circ$$

$$\frac{4000}{\sqrt{3}} = F_{DB} + F_{AD}$$

$$F_{DB} + F_{AD} = 2312.14 \text{ N}$$

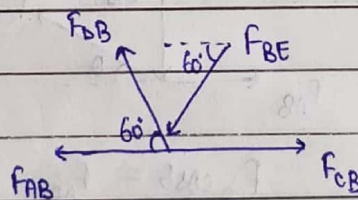
$$F_{DB} = 2312.14 - 2887$$

$$F_{DB} = -574.86 \text{ N (tension)}$$

$$2F_{DE} = -574.86 - 2887$$

$$F_{DE} = -1730.93 \text{ N (compression)}$$

at joint B,



$$\sum F_x = 0;$$

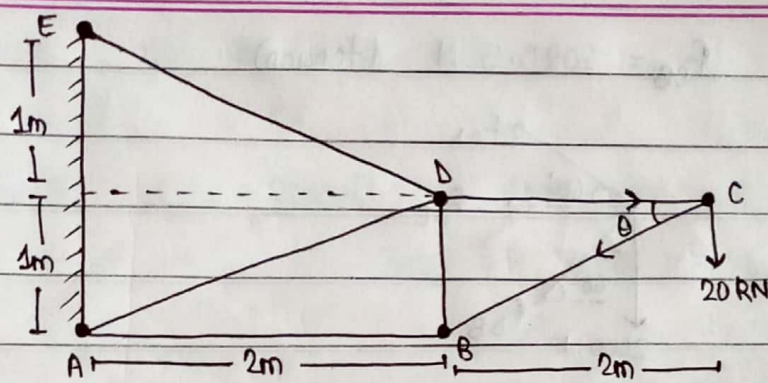
$$F_{BE} \cos 60^\circ + F_{DB} \cos 60^\circ + F_{AB} = F_{CB}$$

$$\frac{F_{BE}}{2} - \frac{574.86}{2} + 1443 = 2020.5$$

$$F_{BE} = 580.14 \text{ N (compression)}$$

* Cantilever truss is the one whose one end is fixed while the other is free.

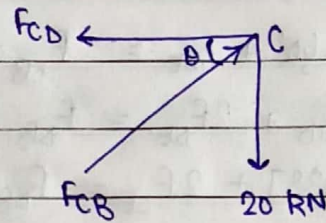
We never calculate reactions at support.



Find the internal forces.

Sol.

at joint C,



(dir.ⁿ can be assumed in either way)

$$\tan \theta = \frac{1}{2} ; \sin \theta = \frac{1}{\sqrt{5}} ; \cos \theta = \frac{2}{\sqrt{5}}$$

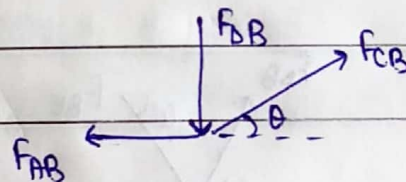
$$\sum F_x = 0 ; \quad F_{CB} \cos \theta = F_{CD}$$

$$\sum F_y = 0 ; \quad 20000 = F_{CB} \sin \theta$$

$$F_{CB} = 20000\sqrt{5} \text{ N (compression)}$$

$$F_{CD} = 40000 \text{ N (tension)}$$

at joint B,



$$\sum F_x = 0 ; \quad F_{CB} \cos \theta = F_{AB}$$

$$F_{AB} = 40000 \text{ N (tension)}$$

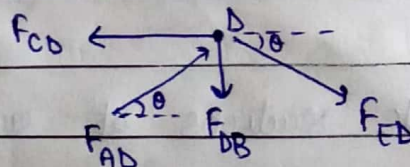
$$\sum F_y = 0 ;$$

$$F_{CB} \sin \theta = F_{DB}$$

$$20000 \text{ N} = F_{DB}$$

(compression)

at joint D,



$$\sum F_x = 0 ;$$

$$F_{CB} = F_{ED} \cos \theta + F_{AB} \cos \theta$$

$$40000 = (F_{ED} + F_{AB}) \cdot \frac{2}{\sqrt{5}}$$

$$F_{ED} + F_{AB} = 20000\sqrt{5} \text{ N}$$

$$\sum F_y = 0 ;$$

$$F_{DB} + F_{ED} \sin \theta = F_{AB} \sin \theta$$

$$20000\sqrt{5} = (F_{AB} - F_{ED})$$

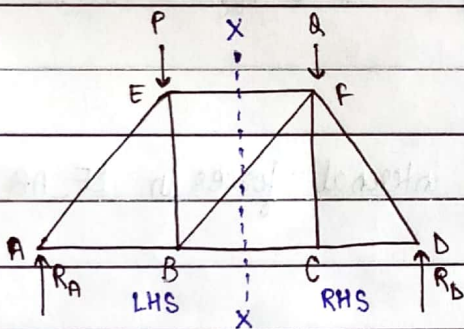
$$20000\sqrt{5} = F_{ED} + F_{AB}$$

$$40000\sqrt{5} = 2F_{AB}$$

$$F_{AB} = 20000\sqrt{5} \text{ N}$$

$$\text{So, } F_{ED} = 0 \text{ N}$$

2. Method of sections



XX $\rightarrow$ section line

Firstly, we divide the truss into 2 equal/unequal parts by drawing a section line.

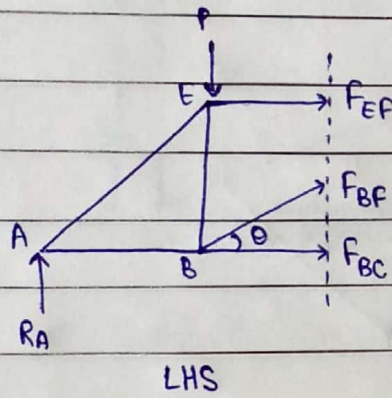
- Section line should not pass through the joints.
- It should pass through that member whose internal force is to be calculated.

- The section line should divide the truss into 2 parts (need not always be equal but separate).
- It should not cut more than 3 members, as we only have 3 equilibrium conditions.

However, in special case, it may cut more than 3 members.

- While applying the moment equation, we can take moment about any point.

To find force in EF:



$$\sum M_B = 0$$

$$F_{EF} \times BE + R_A \times AB = 0$$

$$F_{EF} = -\frac{R_A \times AB}{BE}$$

(F_{EF} is in $\leftarrow$ direction.)

To find force in BC:

$$\sum M_F = 0 ;$$

$$F_{BC} \times CF + P \times BC = R_A \times AC$$

$$F_{BC} \times BE + P \times BC = R_A \times AC$$

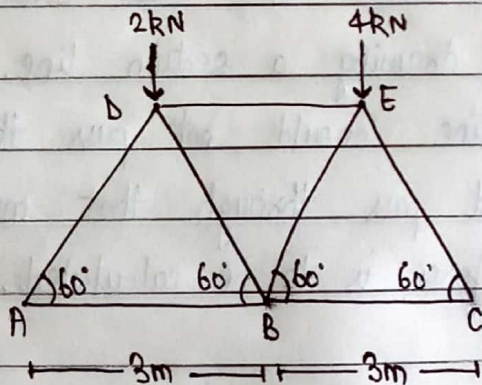
To find force in BF:

$$\sum M_E = 0 ;$$

$$R_A \times AB + F_{BF} \sin \theta \cdot 0 = F_{BF} \cos \theta \cdot BE + F_{BC} \cdot BE$$

$$R_A \times AB = F_{BF} \cos \theta \cdot BE + F_{BC} \cdot BE$$

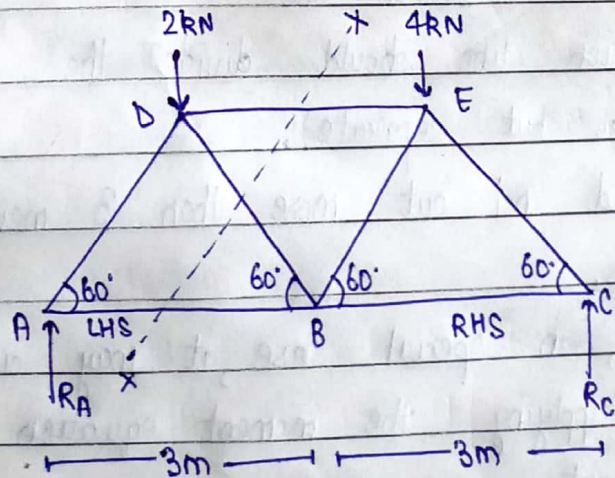
Q.



Find the internal forces in DE, AB and AD.

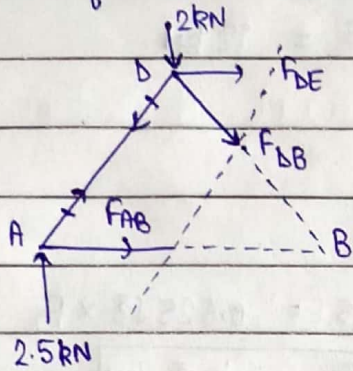
Sol.

Free Body diagram,



By applying method of joints, $R_A = 2.5 \text{ kN}$ and $R_C = 3.5 \text{ kN}$.

To find force in DE :



$$\sum M_B = 0$$

$$2.5 \times 3 + f_{DE} \cdot (1.5\sqrt{3}) = 2(1.5)$$

$$F_{DE} = \frac{3 - 7.5}{1.5\sqrt{3}} = - \frac{4.5}{1.5\sqrt{3}}$$

$$F_{DE} = -\sqrt{3} \text{ N} = -1.73 \text{ kN} \quad (\leftarrow)$$

To find force in AB:

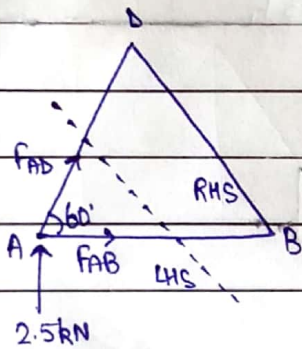
$$\sum M_D = 0 \quad ;$$

$$2.5 \times \sqrt{5} = F_{AB} (\sqrt{5} \sqrt{3})$$

$$F_{AB} = \frac{2.5}{\sqrt{3}} \text{ kN}$$

$$F_{AB} = 1.4 \text{ kN}$$

To find force in AD:



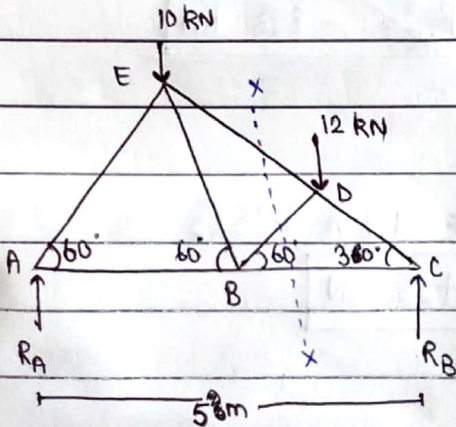
$$\sum M_B = 0$$

$$2.5 \times 8 + F_{AD} \sin 60^\circ \times 8 = 0$$

$$F_{AD} = \frac{-2.5 \times 2}{\sqrt{3}} = -1.66\sqrt{3} \text{ kN} = -2.72 \text{ kN}$$

(F_{AD} is in ↙ direction.)

Q.



Find the internal forces.

Sol.ⁿ

$$\angle BDC = 90^\circ = \angle EDB$$

$$\angle AEB = 60^\circ = \angle EBD$$

$$\angle DEB = 30^\circ$$

$$\angle AED = 90^\circ$$

$$AE = CA \cos 60^\circ = \frac{5 \times 1}{2} = 2.5 \text{ m}$$

$$\sum M_A = 0 \quad ;$$

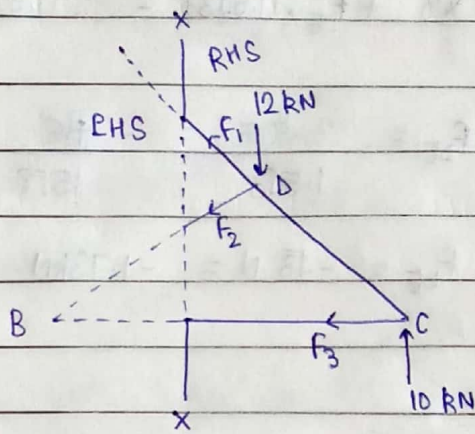
$$R_B \times 5 = 12 \times AE \cos 60^\circ + 12 \times (2.5 + 0.625)$$

$$R_B = \frac{12.5 + 37.5}{5} = 10 \text{ kN}$$

$$\sum M_B = 0 ;$$

$$R_A \times 5 = 12 \times 1.875 + 10 \times 3.75$$

$$R_A = \frac{22.5 + 37.5}{5} = 12 \text{ kN}$$



$$\sum M_D = 0 ;$$

$$10 \times 1.875 = 0.625\sqrt{3} \times F_3$$

$$F_3 = 10\sqrt{3} \text{ kN}$$

$$\sum M_C = 0 ;$$

$$12 \times 1.875 + F_2 \times 1.25\sqrt{3} = 0$$

$$F_2 = \frac{-12 \times 1.875\sqrt{3}}{1.25 \times 3}$$

$$F_2 = -6\sqrt{3} \text{ kN}$$

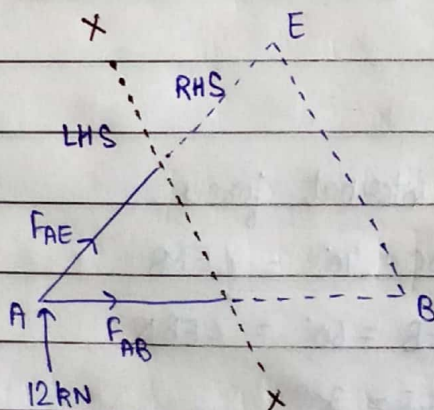
$$\sum M_B = 0 ;$$

$$12 \times 0.675 = 10 \times 2.5 + F_1 \times 0.45\sqrt{3}$$

$$8.1 = 25 + 0.45\sqrt{3} F_1$$

$$\frac{-16.9\sqrt{3}}{0.45 \times 3} = F_1$$

$$F_1 = -12.5\sqrt{3} \text{ kN}$$



$$\sum M_B = 0 ;$$

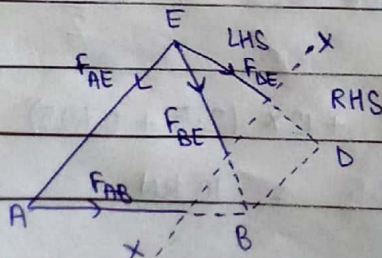
$$12 \times 2.5 + F_{AE} \times 1.25\sqrt{3} = 0$$

$$F_{AE} = -8\sqrt{3} \text{ kN}$$

$$\sum M_E = 0 ;$$

$$12 \times 1.75 = F_{AB} \times 1.75\sqrt{3}$$

$$F_{AB} = 4\sqrt{3} \text{ kN}$$



$$\sum M_B = 0 ;$$

$$F_{AE} \times 1.25\sqrt{3} = F_{DE} \times 1.25$$

$$F_{DE} = -24 \text{ kN}$$

$$\sum M_A = 0 ;$$

$$F_{BE} \times 1.25\sqrt{3} + F_{BE} \times 2.5 = 0$$

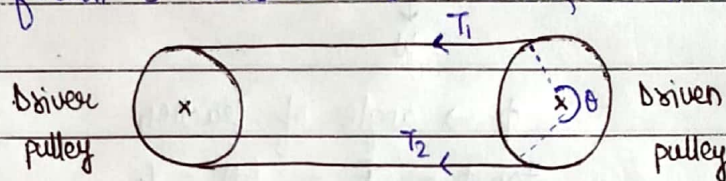
$$F_{BE} = \frac{8 \times 24 \times 2.5 \times \sqrt{3}}{1.25 \times 8} = 16\sqrt{3} \text{ kN}$$

$$F_{BE} = 16\sqrt{3} \text{ kN}$$

Applications of Friction

- 1) Solid friction - between solid surfaces
- 2) Liquid friction - between layers of fluids

Cases where friction is desirable are brakes, clutch and belt or rope drives.



Belt and Rope drive

Belt and rope drive is used to transfer energy from one shaft to the other. If there is no perfect frictional grip between the rope and belt, the tangential velocity of rope and belt would differ and so it won't be able to transfer the energy to the driven pulley.

$\theta \rightarrow$ angle of lap

$r \rightarrow$ radius of pulley

$$T_2 = T_1 + \underbrace{(dF)}_{\text{frictional force}}$$

$$\text{Driven force} = T_2 - T_1$$

$T_2 > T_1$ or $T_1 > T_2$ due to friction.

$$\text{Moment} = (T_2 - T_1) \times r$$

$$\text{Power} = (T_2 - T_1) \times \omega$$

$\hookrightarrow$ angular velocity

Greater is the angle of lap, larger is the difference b/w T_1 and T_2 .

Side where tension is greater $\rightarrow$ tight side

Side where tension is lesser $\rightarrow$ slack side

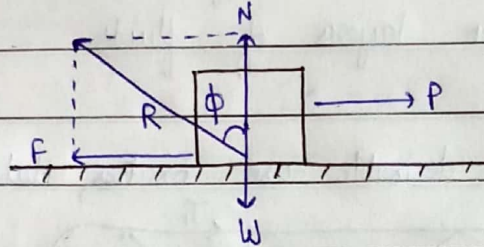
$$(T_2 - T_1) \propto \theta$$

Laws of friction:

- Friction is independent of area of contact.
- Depends on nature of surface (coefficient of friction)

Angle of friction:

Limiting equilibrium: Applied force = Frictional force



$\phi \rightarrow$ angle of friction
$$\tan \phi = \frac{F}{N} = \frac{\mu N}{N} = \mu$$

$$\boxed{\tan \phi = \mu}$$

Screw friction $\rightarrow$ between bolt and nut
(also called self-locking friction)

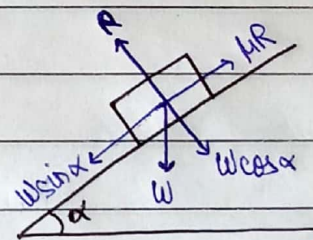
Angle of repose:

$$\mu R = W \sin \alpha$$

$$R = W \cos \alpha$$

$$\mu W \cos \alpha = W \sin \alpha$$

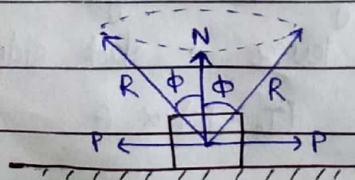
$$\boxed{\tan \alpha = \mu}$$



$\alpha \rightarrow$ angle of repose

Angle of repose is the maximum angle at which the block won't slide down due to its own weight.

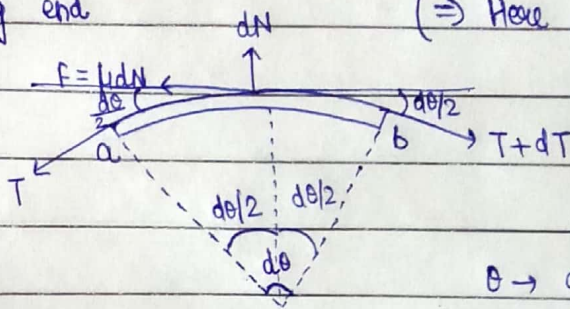
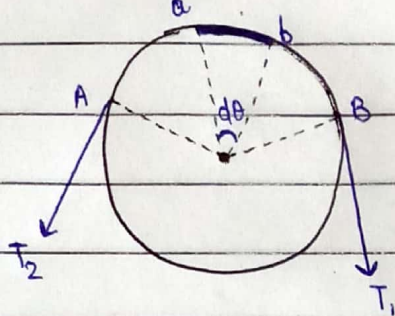
Cone of friction:



A → receiving end

B → leaving end

(⇒ Here $T_1 > T_2$)



θ → angle of lap / angle of contact

$$\sum F_x = 0 ; \quad (T+dT) \cos \frac{d\theta}{2} = T \cos \frac{d\theta}{2} + \mu dN \quad - (1)$$

$$\sum F_y = 0 ; \quad (T+dT) \sin \frac{d\theta}{2} + T \sin \frac{d\theta}{2} = dN \quad - (2)$$

$d\theta$ is very small so $\frac{d\theta}{2} \rightarrow 0 \quad \therefore \cos \frac{d\theta}{2} = 1$ and $\sin \frac{d\theta}{2} \approx \frac{d\theta}{2}$

$$\text{So,} \quad (T+dT) = T + \mu dN$$

$$dT = \mu dN \quad - (3)$$

$$\text{and,} \quad T \frac{d\theta}{2} + \frac{dT(d\theta)}{2} + T \frac{d\theta}{2} = dN$$

$$Td\theta = dN \quad - (4)$$

(∵ dT is very small.)

from eq. (3) and (4)

$$dT = \mu T d\theta$$

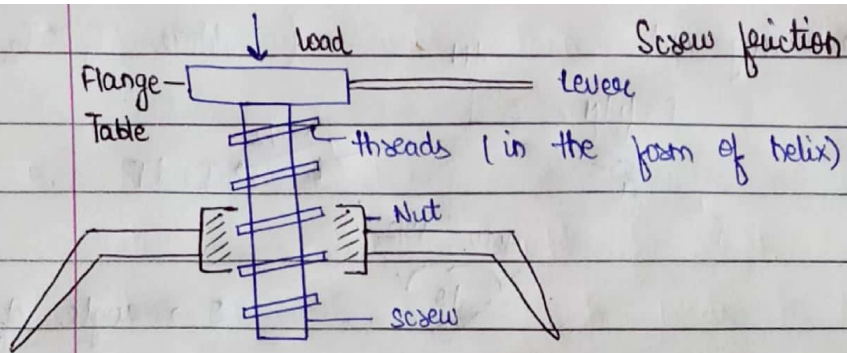
$$\int_{T_2}^{T_1} \frac{dT}{T} = \int_0^\theta \mu d\theta$$

$$[\log T]_{T_2}^{T_1} = \mu \theta$$

$$\log \left(\frac{T_1}{T_2} \right) = \mu \theta$$

$$\boxed{\frac{T_1}{T_2} = e^{\mu \theta}}$$

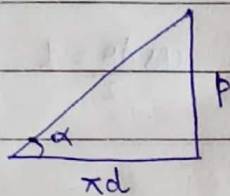
Power transfer increases if angle of lap is more. To increase angle of lap, we use a cross belt over the pulley (as in wheat grinder). Here, friction has +ve application as power increases with increase in μ .



The threads are assumed to be kept on an inclined plane.

So, the load is, in this way, kept on an inclined plane.

Helix angle is the angle of inclined plane.

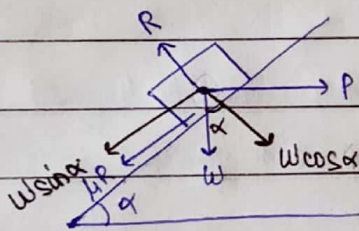


$d \rightarrow$ diameter of screw

$p \rightarrow$ pitch

$$\tan \alpha = \frac{p}{\pi d}$$

$\alpha \rightarrow$ helix angle



P force is applied to lift the load.
in static equilibrium,

$$\sum F_x = 0; \quad P \cos \alpha - \mu R - W \sin \alpha = 0 \quad \text{--- (i)}$$

$$P \cos \alpha = \mu R + W \sin \alpha \quad \text{--- (ii)}$$

$$\sum F_y = 0; \quad R = W \cos \alpha + P \sin \alpha \quad \text{--- (iii)}$$

$$\text{So, } P \cos \alpha = \mu (W \cos \alpha + P \sin \alpha) + W \sin \alpha$$

$$P (\cos \alpha - \mu \sin \alpha) = \mu W \cos \alpha + W \sin \alpha$$

$$P (\cos \alpha - \mu \sin \alpha) = W (\sin \alpha + \mu \cos \alpha)$$

$$P = \frac{W (\sin \alpha + \mu \cos \alpha)}{\cos \alpha - \mu \sin \alpha}$$

$\phi \rightarrow$ angle of friction; $\mu = \tan \phi = \sin \phi / \cos \phi$

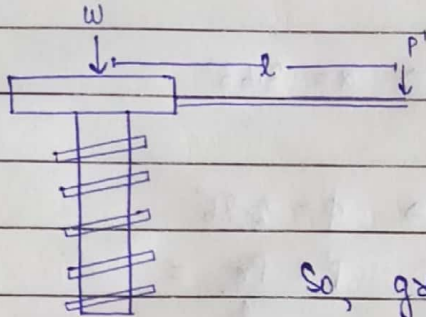
$$P = \frac{W (\sin \alpha + \frac{\sin \phi \cos \alpha}{\cos \phi})}{\cos \alpha - \frac{\sin \phi \sin \alpha}{\cos \phi}}$$

$$P = \frac{W (\sin \alpha \cos \phi + \sin \phi \cos \alpha)}{\cos \alpha \cos \phi - \sin \alpha \sin \phi}$$

$$P = \frac{W(\sin\alpha \cos\phi + \sin\phi \cos\alpha)}{\cos\alpha \cos\phi - \sin\phi \sin\alpha}$$

$$P = \frac{W \sin(\alpha + \phi)}{\cos(\alpha + \phi)}$$

$$P = W \tan(\alpha + \phi)$$



$$P' \times l = P \times r$$

$$P' = \frac{P r}{l}$$

So, greater is the length of the lever, lesser is the effort required to lift the load. This increases the mechanical advantage of the lifting machine.

Screw jack is self locking as the friction b/w the nut and the screw acts upwards and stops the load from falling after it has been lifted to the desired height. For self locking, $\alpha < \phi$.

Mechanical system - mechanical effort as in screw jack

Hydraulic system - hydraulic effort as in cars

Pneumatic system - due to air pressure as in DTC buses

$$\text{Mechanical advantage} = \frac{W}{P} = \frac{\text{Weight of load}}{\text{Force applied}}$$

$$\text{Velocity ratio} = \frac{2\pi r}{p}$$

$$\eta = \text{Efficiency} = \frac{\text{Mechanical advantage}}{\text{Velocity ratio}} = \frac{\text{Ideal effort}}{\text{Actual effort}}$$

$$\eta = \frac{\tan\alpha}{\tan(\alpha + \phi)}$$

η is maximum when

$$\alpha = \frac{\pi}{4} - \frac{\phi}{2}$$

;

$$\eta_{\text{max.}} = \frac{1 - \sin\phi}{1 + \sin\phi}$$

- if load is brought down, $P = W \tan(\alpha - \phi)$

Q. A square threaded screw of 7.5 cm mean diameter and 1.5 cm pitch is utilised to lift a weight of 600 N. Determine

- (a) the tangential force which needs to be applied at the end of a lever of 40 cm ($\mu = 0.05$)

$$\tan \alpha = \frac{P}{\pi d} = \frac{1.5}{3.14 \times 7.5} \Rightarrow \alpha = 3.63^\circ$$

$$\mu = \tan \phi = 0.05 \Rightarrow \phi = 2.86^\circ$$

$$P = W \tan(\alpha + \phi) = ; P' = \frac{P \times l}{r} =$$

- (b) the efficiency of screw jack and state if it is self locking or not.

$$\eta = \frac{\tan \alpha}{\tan(\alpha + \phi)}$$

Since $\alpha > \phi$, it is not self locking.

- (c) the effort to bring down the load

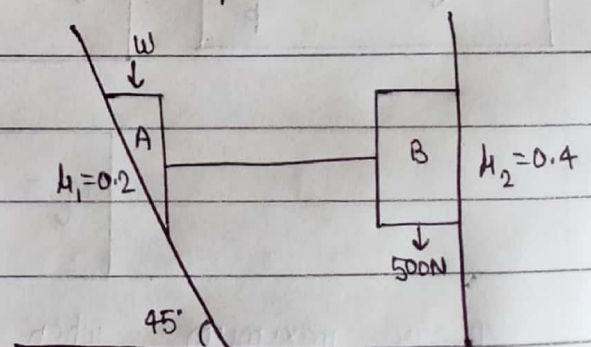
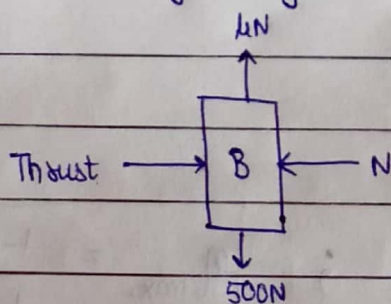
$$P = W \tan(\alpha - \phi)$$

$$P' = \frac{P \times l}{r} =$$

$$P =$$

Q. A block of weight W is placed on a rough inclined plane having coeff. of friction 0.2 and is held in position by means of a horizontal rod hinged to block B, which presses against rough vertical wall having $\mu = 0.4$. The weight of block B is 500 N. Find the min. weight of block A in consistent equilibrium.

Solⁿ Free Body diagram of B:

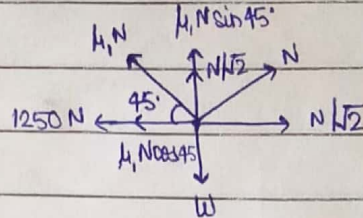
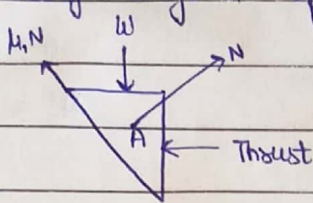


$$\frac{1}{2}N = 500$$

$$N = \frac{500}{0.4} = \frac{5000}{4}$$

$$\text{Thrust} = N = 1250 \text{ N}$$

Free Body diagram of A:



$$\frac{0.2N}{\sqrt{2}} + \frac{N}{\sqrt{2}} = W$$

$$\sqrt{2} W = 1.2 N$$

$$1250 + \frac{0.2N}{\sqrt{2}} = \frac{N}{\sqrt{2}}$$

$$1250\sqrt{2} = 0.8 N$$

$$N = 2209.37 \text{ N}$$

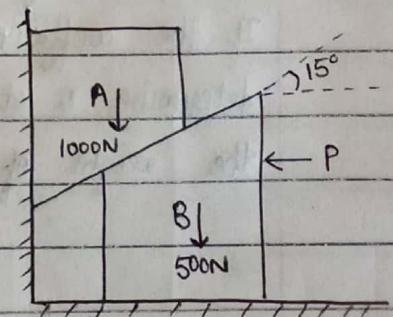
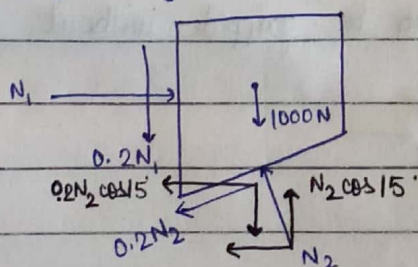
$$\text{So, } W = 1875 \text{ N}$$

Wedge friction

Wedge is an inclined plane used for raising heavy loads.

- Q. A block of weight 1000 N is to be raised by means of a 15° wedge B weighing 500 N. Assuming coeff. of friction for all the contact surfaces to be 0.2, determine the minimum horizontal force P that should be applied to raise the block.

Sol.ⁿ Free body diagram of A:



$$\sum F_x = 0 \quad ; \quad \sum F_y = 0$$

$$N_1 = 0.2N_2 \cos 15^\circ + 0.2N_2 \sin 15^\circ \quad ; \quad 0.2N_1 + 1000 + 0.2N_2 \sin 15^\circ = N_2 \cos 15^\circ$$

$$N_1 = 0.2N_2 (0.96) + N_2 (0.26) \quad ; \quad 0.2N_1 + 1000 + 0.2N_2 (0.26) = N_2 (0.96)$$

$$N_1 = 0.452 N_2 \quad ; \quad 0.2N_1 + 1000 = 0.908 N_2$$

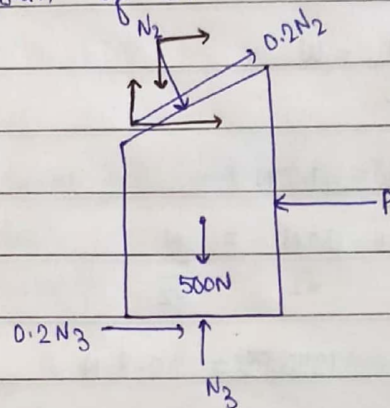
$$N_1 = 552.796 \text{ N}$$

$$0.2(0.452)N_2 + 1000 = 0.908N_2$$

$$1000 = 0.8176 N_2$$

$$N_2 = 1223 \text{ N}$$

Free body diagram of B:



$$\sum F_x = 0 \quad ; \quad 0.2N_2 \cos 15^\circ + 0.2N_3 + N_2 \sin 15^\circ = P$$

$$\sum F_y = 0 \quad ; \quad 500 + N_2 \cos 15^\circ = N_3 + 0.2N_2 \sin 15^\circ$$

$$500 + 1223(0.96) = N_3 + 0.2 \times 1223 \times 0.26$$

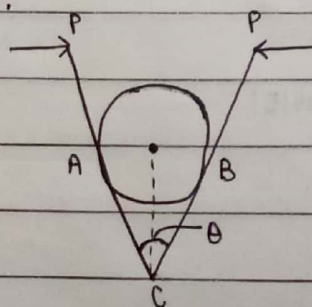
$$N_3 = 1610.484 \text{ N}$$

$$\text{So, } P = 0.2 \times 1223 \times 0.96 + 0.2 \times 1610.484 + 1223 \times 0.26$$

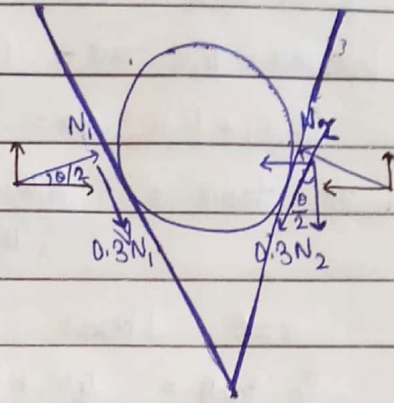
$$P = 234.816 + 322.0968 + 317.98$$

$$P = 874.89 \text{ N}$$

- Q. A pipe is gripped between the 2 levers that are pinned together at C. If the coefficient of friction between the levers and pipes is 0.3, determine θ at which the pipe can be gripped without slip. Neglect the weight of pipe.



Sol.ⁿ Free body diagram of pipe:



$\sum F_x = 0$; All components get cancelled, and we get $N_1 = N_2$.

$\sum F_y = 0$; $N_1 = N_2 = N$ (say) (due to symmetry)

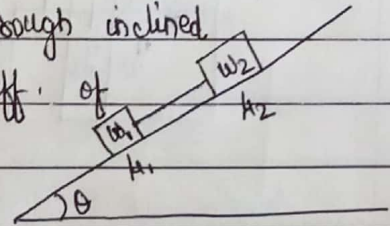
$$\frac{2 \times N \sin \theta}{2} = \frac{2 \times 0.3 N \cos \theta}{2}$$

$$\tan \theta = 0.3$$

$$\theta = 2 \tan^{-1}(0.3)$$

When the 2 levers would approach each other, the tendency of pipe to move would be in upward dir.ⁿ so friction acts in downward direction.

a. 2 blocks of weight w_1 and w_2 rest on a rough inclined plane and are connected by a string. The coeff. of friction b/w these blocks are μ_1 and μ_2 respectively so that the blocks



would lie on the point of contact when the inclination of the plane is given by $\tan \theta = \frac{\mu_1 w_1 + \mu_2 w_2}{w_1 + w_2}$, what will be the inclination when

$w_1 = w_2$ and $\mu_1 = \frac{1}{2}$ and $\mu_2 = \frac{1}{3}$.

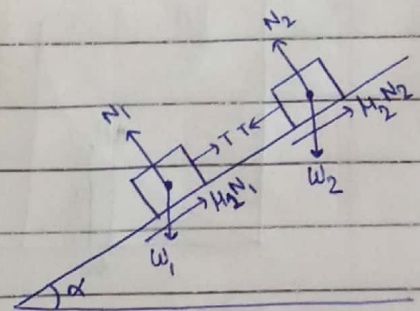
Sol.ⁿ for w_1 , $\sum F_x = 0$; $T + \mu_1 N_1 = w_1 \sin \alpha$ — (i)

$$\sum F_y = 0 ; N_1 = w_1 \cos \alpha$$

for w_2 , $\sum F_x = 0$; $T + w_2 \sin \alpha = \mu_2 N_2$

$$\sum F_y = 0 ; N_2 = w_2 \cos \alpha$$

$$\text{or } \mu_2 N_2 = T + w_2 \sin \alpha \text{ — (ii)}$$



$$\text{eq. (I)} + \text{eq. (II)}$$

$$\mu_1 N_1 + \mu_2 N_2 = W_1 \sin \alpha + W_2 \sin \alpha$$

$$\mu_1 W_1 \cos \alpha + \mu_2 W_2 \cos \alpha = (W_1 + W_2) \sin \alpha$$

$$(\mu_1 W_1 + \mu_2 W_2) \cos \alpha = (W_1 + W_2) \sin \alpha$$

$$\Rightarrow \tan \alpha = \frac{\mu_1 W_1 + \mu_2 W_2}{W_1 + W_2}$$

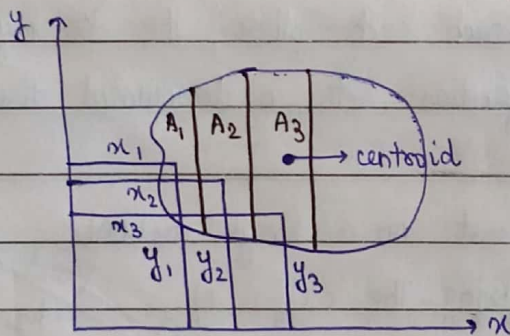
$$\alpha = \theta \quad (\text{here})$$

$$\text{So, } \tan \theta = \frac{\mu_1 W_1 + \mu_2 W_2}{W_1 + W_2}$$

$$\text{if } W_1 = W_2 ; \mu_1 = \frac{1}{2} \text{ and } \mu_2 = \frac{1}{3},$$

$$\tan \theta = \frac{\frac{W_1}{2} + \frac{W_1}{3}}{2W_1} = \frac{5}{12}$$

Centre of Area / Centroid



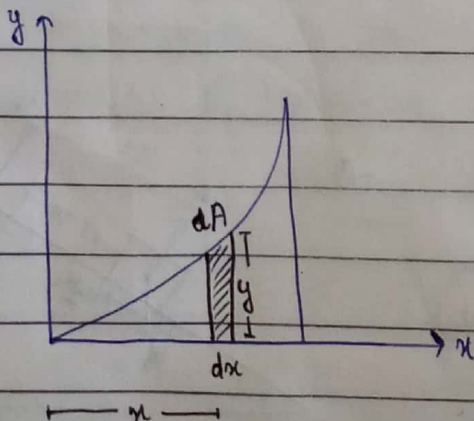
$A \rightarrow$ Total area.

$$A \bar{x} = A_1 x_1 + A_2 x_2 + A_3 x_3$$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3}{A_1 + A_2 + A_3}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3}$$

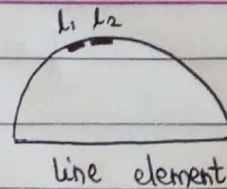
$$\text{So, } \bar{x} = \frac{\sum A_i x_i}{\sum A_i} ; \bar{y} = \frac{\sum A_i y_i}{\sum A_i} \quad (\text{for regular shapes})$$



$$dA = y dx$$

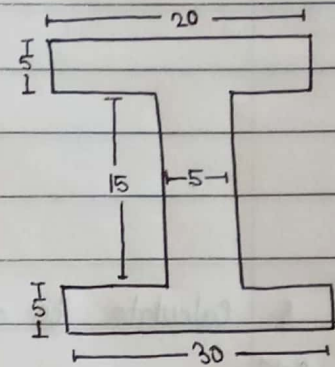
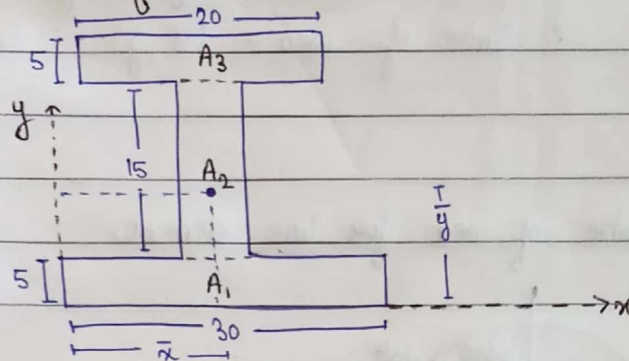
$$\bar{y} = \frac{\int (y dx) \cdot \frac{y}{2}}{\int y dx} \quad (\text{for irregular shape})$$

for a line element: $\bar{y} = \frac{\sum l_i y_i}{\sum l_i}$



Q. Calculate the centre of area.

Sol.ⁿ



NOTE: Centroid always lies on the axis about which the given section is symmetrical.

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3}$$

$$A_1 = 150 \text{ cm}^2 ; y_1 = 2.5 \text{ cm}$$

$$A_3 = 100 \text{ cm}^2 ; y_3 = 22.5 \text{ cm}$$

$$A_2 = 75 \text{ cm}^2 ; y_2 = 12.5 \text{ cm}$$

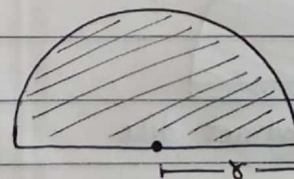
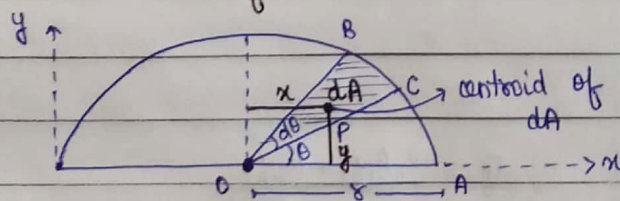
$\bar{x} = 15 \text{ cm}$ | as the section is symmetrical about y-axis

$$\bar{y} = \frac{150 \times 2.5 + 75 \times 12.5 + 100 \times 22.5}{325} = 10.96 \text{ cm}$$

$$\therefore \text{Centre of area} = (\bar{x}, \bar{y}) = (15, 10.96)$$

Q. Calculate the centre of area.

Sol.ⁿ



$$BC = x \sin \theta$$

$$\text{So, } dA = \frac{1}{2} x \sin \theta \times x = \frac{1}{2} x^2 \sin \theta ; OP = \frac{2x}{3} (\because OBC \text{ is a } \Delta)$$

$$y = \frac{2x}{3} \sin \theta ; x = \frac{2x}{3} \cos \theta$$

$$\bar{y} = \frac{\int y dA}{\int dA}$$

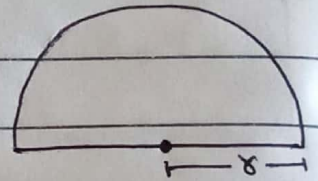
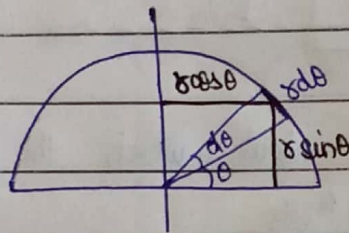
$$\bar{y} = \frac{\int_0^\pi \frac{2}{3} r \sin \theta \times \frac{1}{2} r^2 d\theta}{\int_0^\pi \frac{1}{2} r^2 d\theta} = \frac{4r}{3\pi}$$

$\bar{x} = r$ (due to symmetry)

$\therefore$ Centre of area = $(\bar{x}, \bar{y}) = (r, \frac{4r}{3\pi})$

Q. Calculate the centre of area for line element.

Soln

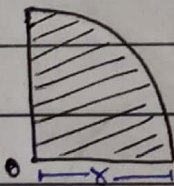


$$\bar{y} = \frac{\int_0^\pi r d\theta \cdot r \sin \theta}{\int_0^\pi r d\theta} = \frac{2r}{\pi}$$

$\bar{x} = 0$ (due to symmetry)

$\therefore$ Centre of area = $(\bar{x}, \bar{y}) = (0, \frac{2r}{\pi})$

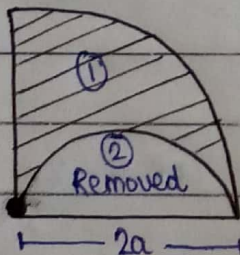
NOTE: (i)



Quadrant disc

Centre of area = $(\frac{4r}{3\pi}, \frac{4r}{3\pi})$

(ii)



$$\bar{y} = \frac{A_1 y_1 - A_2 y_2}{A_1 - A_2}$$

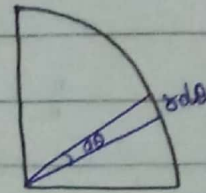
$$\bar{y} = \frac{A_1 \frac{4(2a)}{3\pi} - A_2 \frac{4a}{3\pi}}{A_1 - A_2}$$

Q. Calculate the centre of area for line element.

Sol.ⁿ

$$\bar{y} = \frac{\int_0^{x/2} x d\theta \cdot x \sin \theta}{\int_0^{x/2} x d\theta} = \frac{2x}{x} = \bar{x}$$

$$\therefore \text{Centre of area} = (\bar{x}, \bar{y}) = \left(\frac{2x}{x}, \frac{2x}{x} \right)$$



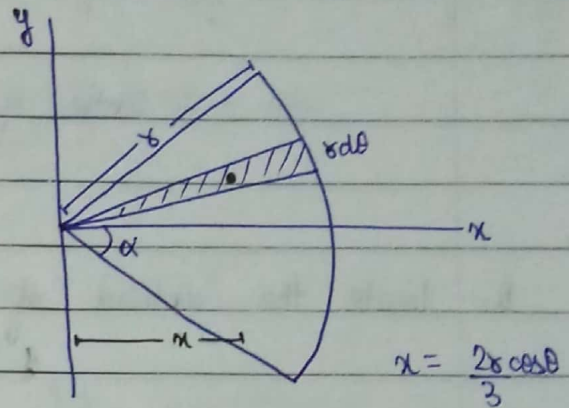
For a sector of a circle:

$$dA = \frac{1}{2} x d\theta \times x$$

$$\bar{x} = \frac{\int dA \cdot x}{\int dA}$$

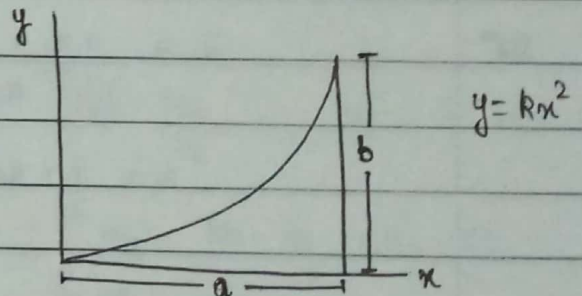
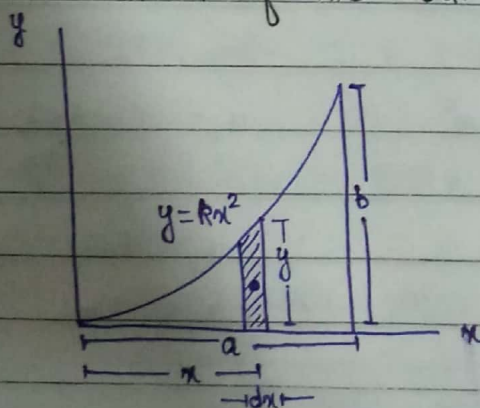
$$\bar{x} = \frac{\int \frac{2x}{3} \cos \theta \cdot \frac{1}{2} x^2 d\theta}{\int \frac{1}{2} x^2 d\theta}$$

$$= \frac{2x(\sin \theta)}{3 \theta}$$



Q. Find the centroid of this area.

Sol.ⁿ



$$y = \frac{b}{a^2} x^2$$

$$y = kx^2$$

$$x = a ; y = b$$

$$b = ka^2 \Rightarrow k = \frac{b}{a^2}$$

$$\text{So, } y = \frac{b}{a^2} x^2$$

$$dA = y dx$$

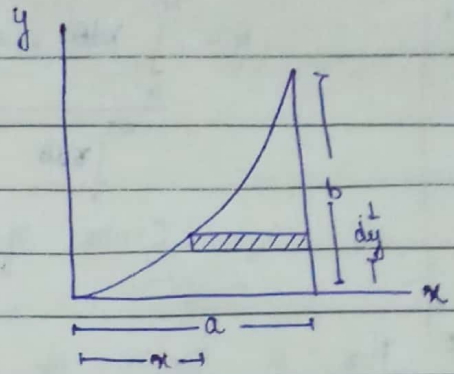
$$\bar{y} = \frac{\int_0^a y dx \cdot \frac{y}{2}}{\int_0^a y dx} = \frac{3}{10} b$$

$$\bar{x} = \frac{\int_0^a y dx \cdot x}{\int_0^a y dx} = \frac{3}{4} a$$

OR

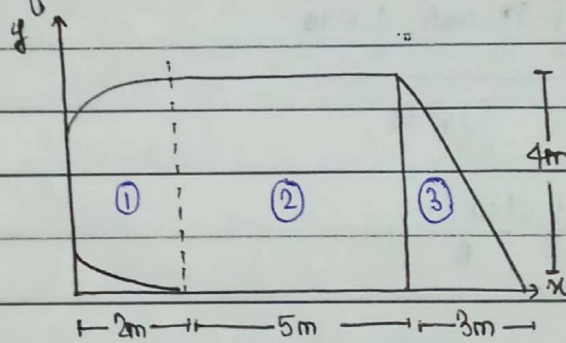
$$\bar{y} = \frac{\int_0^b (a-x) dy \cdot y}{\int_0^b (a-x) dy} = \frac{3b}{10}$$

$$\bar{x} = \frac{\int_0^b (a-x) dy \cdot x}{\int_0^b (a-x) dy} = \frac{3a}{4}$$



$$\therefore \text{Centre of area} = \left(\frac{3a}{4}, \frac{3b}{10} \right)$$

Q. Locate the centroid of the area w.r.t axis shown in figure.



Soln

$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3}{a_1 + a_2 + a_3}$$

$$a_1 = \frac{1}{2} \times \pi \times 4 = 2\pi \text{ m}^2$$

$$x_1 = 2 - \frac{4 \times 2}{3\pi} = 2 - \frac{8}{3\pi}$$

$$a_2 = 20 \text{ m}^2$$

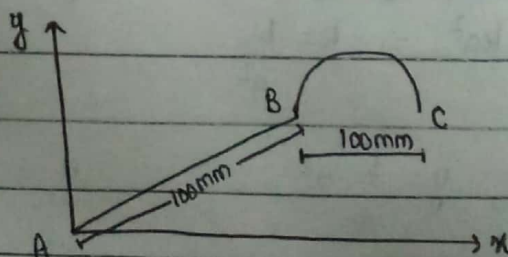
$$x_2 = 4.5$$

$$a_3 = \frac{1}{2} \times 3 \times 4 = 6 \text{ m}^2$$

$$x_3 = 8 \text{ m}$$

$$\bar{x} = \frac{2\pi \left(2 - \frac{8}{3\pi} \right) + 90 + 48}{2\pi + 26}$$

Q. Locate the centroid.



Solⁿ

$$\bar{x} = \frac{l_1 x_1 + l_2 x_2}{l_1 + l_2}$$

$$l_1 = 100 \text{ mm}$$

$$x_1 = 50 \cos 30^\circ = 25\sqrt{3}$$

$$l_2 = \pi x = 50\pi \text{ mm}$$

$$x_2 = 100 \cos 30^\circ + 50 = 50(\sqrt{3} + 1)$$

$$\bar{x} = \frac{100 \times 25\sqrt{3} + 50\pi \times 50(\sqrt{3} + 1)}{100 + 50\pi} = 100.3 \text{ mm}$$

$$y_1 = 50 \sin 30^\circ = 25$$

$$y_2 = \frac{2x}{\pi} + 100 \sin 30^\circ = 50 + \frac{2x}{\pi}$$

$$\bar{y} = \frac{100 \times 25 + 50\pi \left(50 + \frac{2x}{\pi}\right)}{100 + 50\pi} = 59.4 \text{ mm}$$

$\therefore$ Centre of area = (100.3, 59.4)

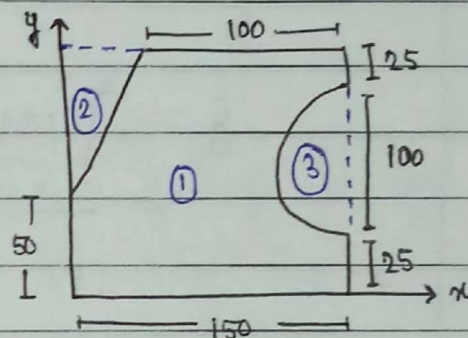
Q. Locate the centroid of the given area.

Solⁿ

$$\bar{x} = \frac{a_1 x_1 - a_2 x_2 - a_3 x_3}{a_1 - a_2 - a_3}$$

$$a_1 = 22500 \text{ mm}^2 ; x_1 = 75 \text{ mm}$$

$$a_2 = \frac{1}{2} \times 50 \times 100$$

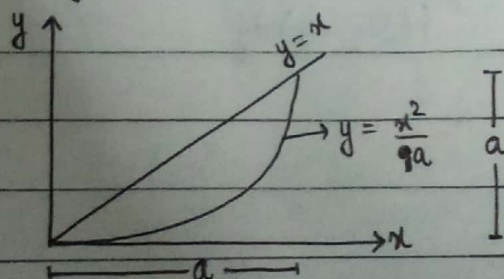


$$a_2 = 2500 \text{ mm}^2 ; x_2 = \frac{150 - \frac{4x}{3}}{3\pi} = 150 - \frac{200}{3\pi} ; y_2 = \frac{50}{3} \text{ mm}$$

$$a_3 = \frac{1}{2} \pi \times 2500 = 1250\pi \text{ mm}^2 ; x_3 = \frac{150 - \frac{4x}{3}}{3\pi} = 150 - \frac{200}{\pi}$$

$$\bar{x} = \frac{75 \times 22500 - 2500 \times \frac{50}{3} - 1250\pi \left(150 - \frac{200}{\pi}\right)}{22500 - 2500 - 1250\pi}$$

Q. Locate the centroid of the given area.



Sol:

$$dA = \left(x - \frac{x^2}{a}\right) dx$$

$$\bar{x} = \frac{\int_0^a x \left(x - \frac{x^2}{a}\right) dx}{\int_0^a \left(x - \frac{x^2}{a}\right) dx}$$

$$\bar{x} = \frac{\left[\frac{x^3}{3} - \frac{x^4}{4a}\right]_0^a}{\left[\frac{x^2}{2} - \frac{x^3}{3a}\right]_0^a} = \frac{\frac{a^3}{3} - \frac{a^3}{4}}{\frac{a^2}{2} - \frac{a^2}{3}} = \frac{\frac{a^3}{12}}{\frac{a^2}{6}} = \frac{a}{2}$$

$$\bar{y} = \frac{\int_0^a y \left(x - \frac{x^2}{a}\right) dx}{\int_0^a \left(x - \frac{x^2}{a}\right) dx}$$

$$y = y_1 + \frac{y_2 - y_1}{2} = \frac{y_1 + y_2}{2}$$

$$y_1 = \frac{x^2}{a}; y_2 = x$$

$$\bar{y} = \frac{\int_0^a \left(\frac{x + \frac{x^2}{a}}{2}\right) \left(x - \frac{x^2}{a}\right) dx}{\frac{a^2}{6}} = \frac{3}{a^2} \int_0^a \left(x^2 - \frac{x^4}{a^2}\right) dx$$

$$\bar{y} = \frac{3}{a^2} \left[\frac{x^3}{3} - \frac{x^5}{5a^2} \right]_0^a = \frac{3}{a^2} \left[\frac{a^3}{3} - \frac{a^3}{5} \right] = \frac{3}{a^2} \cdot \frac{2a^3}{15} = \frac{2a}{5}$$

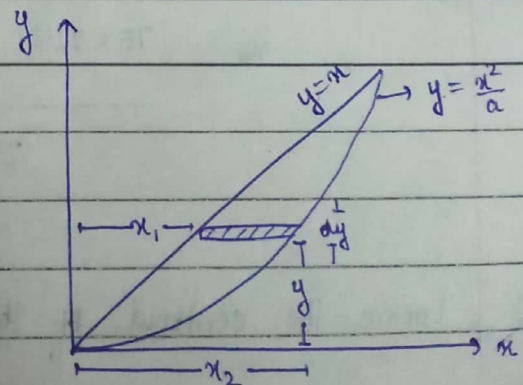
$$\therefore \text{Centroid} = \left(\frac{a}{2}, \frac{2a}{5}\right)$$

OR

$$dA = (x_2 - x_1) dy$$

$$dA = (\sqrt{ay} - y) dy$$

$$\bar{x} = \frac{\int_0^a x (\sqrt{ay} - y) dy}{\int_0^a (\sqrt{ay} - y) dy}$$



$$x = \frac{x_1 + x_2}{2}$$

$$x = \frac{\sqrt{ay} + y}{2}$$

$$\bar{x} = \frac{\int_0^a (\sqrt{ay} + y)(\sqrt{ay} - y) dy}{\int_0^a (\sqrt{ay} - y) dy}$$

$$\bar{x} = \frac{\frac{1}{2} \int_0^a (ay - y^2) dy}{\left[\frac{2\sqrt{a}}{3} y^{3/2} - \frac{y^2}{2} \right]_0^a} = \frac{\frac{1}{2} \left[\frac{ay^2}{2} - \frac{y^3}{3} \right]_0^a}{\frac{2}{3} a^2 - \frac{a^2}{2}}$$

$$\bar{x} = \frac{\frac{1}{2} \left[\frac{a^3}{2} - \frac{a^3}{3} \right]}{\frac{a^2}{6}} = \frac{\frac{a^3}{6}}{\frac{a^2}{6}} = \frac{a}{2}$$

$$\bar{y} = \frac{\int_0^a y(\sqrt{ay} - y) dy}{\int_0^a (\sqrt{ay} - y) dy} = \frac{\int_0^a (\sqrt{a} y^{3/2} - y^2) dy}{\frac{a^2}{6}}$$

$$\bar{y} = \frac{\left[\frac{2\sqrt{a}}{5} y^{5/2} - \frac{y^3}{3} \right]_0^a}{\frac{a^2}{6}} = \frac{\frac{2}{5} a^3 - \frac{a^3}{3}}{\frac{a^2}{6}} = \frac{2a}{5}$$

$$\therefore \text{Centroid} = \left(\frac{a}{2}, \frac{2a}{5} \right)$$

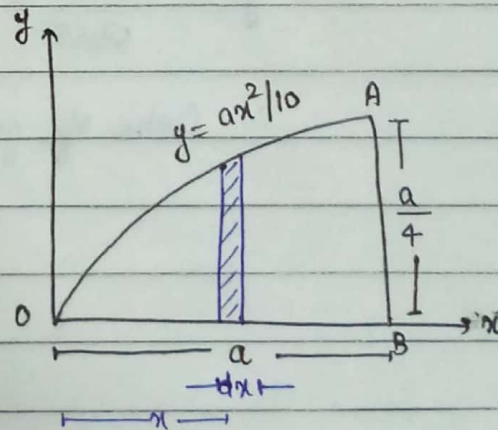
A. Locate the centre of area.

Sol. $dA = \frac{ax^2 dx}{10}$

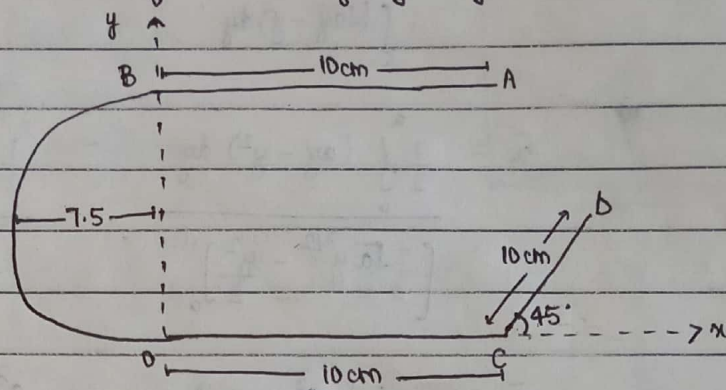
$$\bar{x} = \frac{\int x \cdot \frac{ax^2}{10} dx}{\int \frac{ax^2}{10} dx}$$

$$\bar{x} = \frac{\left[\frac{ax^4}{40} \right]_0^a}{\left[\frac{ax^3}{30} \right]_0^a} = \frac{\frac{a^5}{40}}{\frac{a^4}{30}} = \frac{3a}{4}$$

$$\bar{y} = \frac{\int y \cdot \frac{ax^2}{10} dx}{\int \frac{ax^2}{10} dx} = \frac{\int \frac{ax^2}{20} \cdot \frac{ax^2}{10} dx}{\frac{a^4}{30}} = \frac{\left[\frac{a^2}{200} \cdot \frac{x^5}{5} \right]_0^a}{\frac{a^4}{30}} = \frac{30a^3}{1000} = \frac{3a^3}{100}$$



- Q. A uniform wire is bent into the shape as shown in figure. Calculate the position of centre of gravity w.r.t the given axis.



Sol.

$$\bar{x} = \frac{l_1 x_1 + l_2 x_2 + l_3 x_3 + l_4 x_4}{l_1 + l_2 + l_3 + l_4}$$

$$\bar{x} = \frac{10 \times 5 + (-7.5\pi \times \frac{2 \times 7.5}{\pi}) + 10 \times 5 + 10 \times (10 + 5 \cos 45^\circ)}{10 + 7.5\pi + 10 + 10}$$

$$\bar{x} = 2.29$$

$$\bar{y} = \frac{l_1 y_1 + l_2 y_2 + l_3 y_3 + l_4 y_4}{l_1 + l_2 + l_3 + l_4}$$

$$= \frac{10 \times 15 - 7.5\pi \times 7.5 + 10 \times 0 + 10 \times 5 \sin 45^\circ}{10 + 7.5\pi + 10 + 10}$$

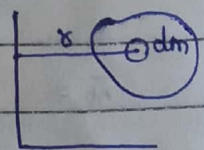
$$\bar{y} = \frac{8.73}{53.55} = 0.163$$

$$\therefore \text{Centre of gravity} = (2.29, 0.163)$$

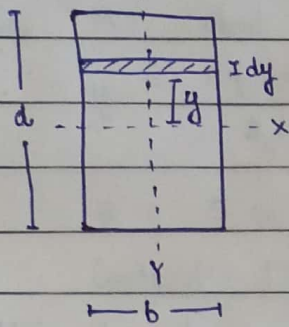
Moment of Inertia

Load taking capacity depends upon second moment of inertia.

$$I = \sum Mx^2 = Ax^2$$



Axes which pass through the centre of gravity are called centroidal axes.



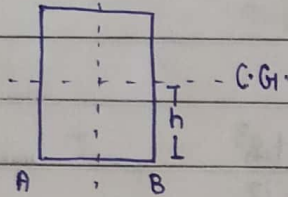
$$dA = b dy$$

$$dI_{xx} = b dy \cdot y^2$$

$$I_{xx} = \int_{-d/2}^{d/2} b dy \cdot y^2 = \frac{bd^3}{12}$$

$$\text{Similarly, } I_{yy} = \frac{db^3}{12}$$

÷ Parallel Axes Theorem:

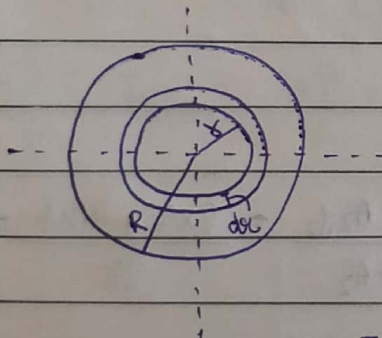


$$I_{AB} = I_{CG} + A x^2$$

$$= \frac{bd^3}{12} + bd \times \left(\frac{d}{2}\right)^2$$

$$I_{AB} = \frac{bd^3}{3}$$

• Inertia about Z-axis is called polar moment of inertia.



$$dA = 2\pi x \cdot dx$$

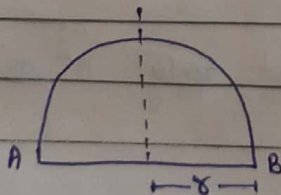
$$dI_z = 2\pi x \cdot dx \cdot x^2$$

$$I_{xx} = \int_0^R 2\pi x \cdot dx \cdot x^2 = \frac{2\pi x^4}{4} \Big|_0^R = \frac{2\pi R^4}{4} = \frac{\pi R^4}{2}$$

$$I_{zc} = I_{xx} + I_{yy}$$

$$2I_{xx} = \frac{\pi R^4}{2} \Rightarrow I_{xx} = \frac{\pi R^4}{4}$$

$$I_{yy} = \frac{\pi R^4}{4}$$



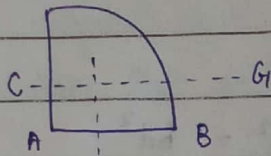
$$I_{AB} = \frac{\pi r^4}{8}$$

Applying 11th axis theorem,

$$I_{AB} = I_{CG} + A h^2$$

$$\frac{\pi r^4}{8} = I_{CG} + \frac{\pi r^2}{2} \cdot \frac{16r^2}{9\pi^2}$$

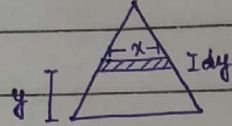
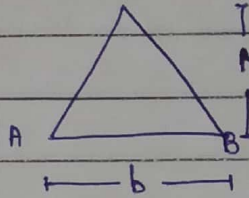
$$I_{CG} = \frac{\pi r^4}{8} - \frac{8r^4}{9\pi}$$



$$I_{AB} = \frac{\pi r^4}{16}$$

$$I_{AB} = I_{CG} + Ab^2$$

$$I_{CG} = 0.55 r^4$$



Using property of similar Δ s,

$$\frac{x}{b} = \frac{h-y}{h} \Rightarrow x = \frac{b(h-y)}{h}$$

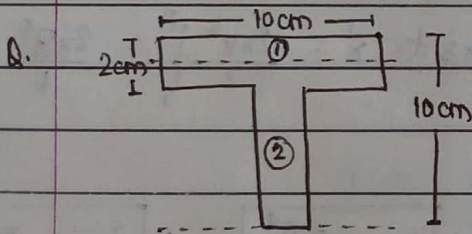
$$dA = \frac{b(h-y)dy}{h}$$

$$I_{BC} = \int \frac{b(h-y)dy}{h} \cdot y^2 = \frac{bh^3}{12}$$

$$I_{BC} = I_{CG} + Ab^2$$

$$\frac{bh^3}{12} = I_{CG} + \frac{1}{2}bh \times \frac{h^2}{9} \Rightarrow I_{CG} = \frac{bh^3}{12} - \frac{bh^3}{18} = \frac{3-2}{36}$$

$$I_{CG} = \frac{bh^3}{36}$$



$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \frac{8 \times 9 + 16 \times 4}{24} = \frac{72 + 64}{24}$$

$$\bar{y} = \frac{138}{24} = 6.77 \text{ cm}$$

Solⁿ $I_{xx} = I_{xx_1} + I_{xx_2}$

We first calculate M.I. about that fig. and apply 11^{th} axis theorem,

$$MI_{CG} = \frac{bd^3}{12} = \frac{10 \times 2^3}{12}$$

$$I_{xx_1} = I_{CG_1} + Ah_1^2 = \frac{10 \times 2^3}{12} + 20 \times (2.23)^2 = 106.125 \text{ cm}$$

$$I_{CG_1} = \frac{db^3}{12} = \frac{8 \times 2^3}{12}; I_{xx_2} = \frac{16}{3} + 16 \times (2.77)^2 = 128.0664$$

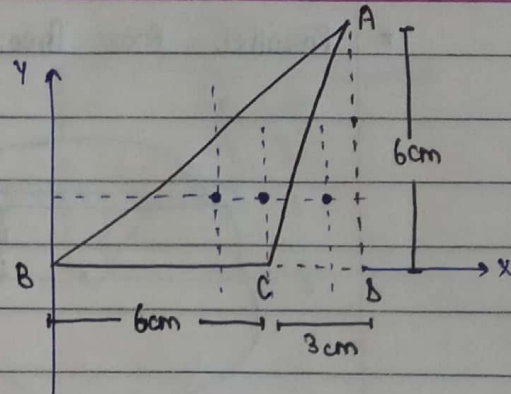
$$I_{xx} = I_{xx_1} + I_{xx_2} = 234.191$$

Q. Find M.I. about the centroidal axis.

Sol.ⁿ $\bar{y} = \frac{A_1 y_1 - A_2 y_2}{A_1 - A_2} = 2 \text{ cm}$

$\bar{x} = 5 \text{ cm}$

Centroid = $(\bar{x}, \bar{y}) = (5, 2)$



$I_{xx} (ABC) = I_{xx_1} (ABD) - I_{xx_2} (ABC)$

$I_{xx_1} = I_{CG_1} + A_1 h_1^2$
 $= \frac{9 \times 6^3}{12} \text{ cm}^4$

$[\because h_1 = 0]$

$I_{xx_2} = \frac{3 \times 6^3}{36} \text{ cm}^4$

$I_{yy} = I_{yy_1} - I_{yy_2}$

$I_{yy_1} = I_{CG_1} + A_1 h_1^2$
 $= \frac{6 \times 9^3}{12} + \frac{1}{2} \times 9 \times 6 \times 1^2 = 148.5$

$[\because h_1 = 6 - 5 = 1]$

$I_{yy_2} = \frac{6 \times 3^3}{36} + \frac{1}{2} \times 3 \times 6 \times 3^2 = 88.5$

$[\because h_2 = 8 - 5 = 3]$

$I_{yy} = 63$

Q. Find M.I. about the centroidal axis.

Sol.ⁿ Centroid = $(\bar{x}, \bar{y}) = (6, 7.5)$

$I_{xx} = I_{xx_1} (\text{Rectangle}) - 2I_{xx_2} (\text{Semi-circle})$

$I_{xx_1} = \frac{12 \times 15^3}{12} \text{ cm}^4$

$I_{xx_2} = \frac{2\pi r^4}{8} \text{ cm}^4 = \frac{\pi r^4}{4} \text{ cm}^4$

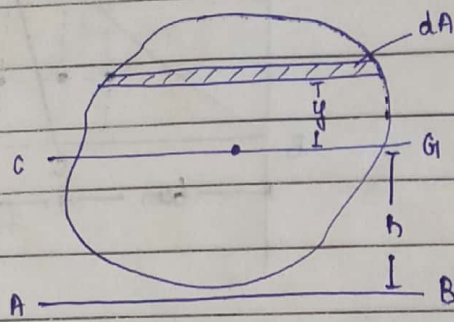
$I_{yy} = I_{yy_1} - 2I_{yy_2}$

$I_{yy_1} = \frac{15 \times 12^3}{12} \text{ cm}^4$

$I_{yy_2} = I_{CG_1} + A_1 h_1^2 = 0.11 \times 5^4 + \frac{\pi \times 25}{2} \left(6 - \frac{4 \times 5}{3\pi}\right)^2$



* Parallel Axis Theorem:



$$dI_{CG} = dA y^2$$

$$I_{CG} = \sum dA y^2$$

$$I_{CG} = \sum dA (b^2 + y^2 + 2by)$$

$$= \sum dA b^2 + \sum dA y^2 + 2b \sum dA y$$

$$= \sum dA b^2 + \bar{I}_{CG} \quad \rightarrow \text{I moment} = 0$$

$$I_{CG} = \bar{I}_{CG} + \sum b^2 dA$$

Q. Find the moment of inertia.

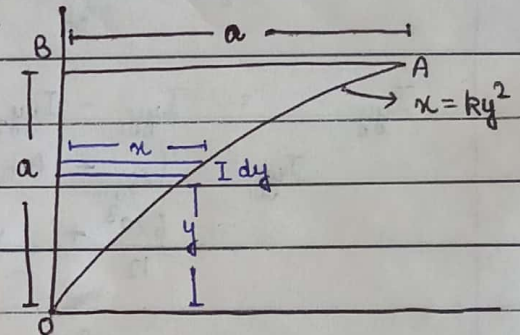
Sol.ⁿ

$$x = ky^2$$

$$\text{at } (a, a)$$

$$a = ka^2 \Rightarrow k = \frac{1}{a}$$

$$\text{So, } \boxed{x = \frac{y^2}{a}}$$



$$dA = x dy = \frac{y^2}{a} dy$$

$$dI_{ox} = \frac{y^2}{a} dy \cdot y^2$$

$$I_{ox} = \int_0^a \frac{1}{a} y^4 dy = \frac{a^5}{5a} = \frac{a^4}{5}$$

$$dI_{oy} = dA \cdot \left(\frac{x}{2}\right)^2 = \frac{y^2}{a} dy \cdot \frac{x^2}{4} = \frac{1}{4a} \cdot \frac{y^6}{a^2} dy = \frac{y^6}{4a^3} dy$$

$$I_{oy} = \int_0^a \frac{y^6}{4a^3} dy = \frac{a^7}{28a^3} = \frac{a^4}{28}$$

$$dA = (y_2 - y_1) dx = (a - \sqrt{ax}) dx$$

$$dI_{ox} = dA \cdot \left(y_1 + \frac{y_2 - y_1}{2} \right) = dA \left(\frac{y_1 + y_2}{2} \right)$$

$$\int dI_{ox} = \int_0^a (a - \sqrt{ax}) \left(\frac{\sqrt{ax} + a}{2} \right) dx$$

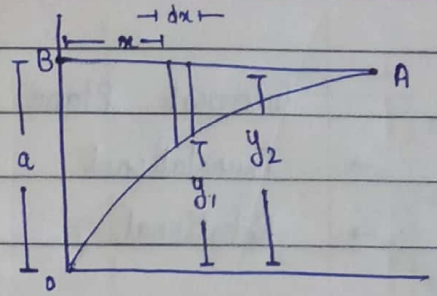
$$= \int_0^a \left(\frac{a^2 - ax}{2} \right) dx \cdot (\sqrt{ax} + a)$$

$$I_{ox} = \frac{1}{2} \left[\frac{a^2 x}{2} + \frac{ax^2}{2} \right]_0^a = \frac{1}{2} \left[\frac{a^3}{2} + \frac{a^3}{2} \right]$$

$$= \frac{1}{(2)^2} \left[a^{5/2} \sqrt{x} + a^3 - (ax)^{3/2} - a^2 x \right]_0^a$$

$$= \frac{1}{(2)^2} \left[\frac{2a^{5/2} x^{3/2}}{3} + a^3 x - \frac{2a^{3/2} x^{5/2}}{5} - \frac{a^2 x^2}{2} \right]_0^a$$

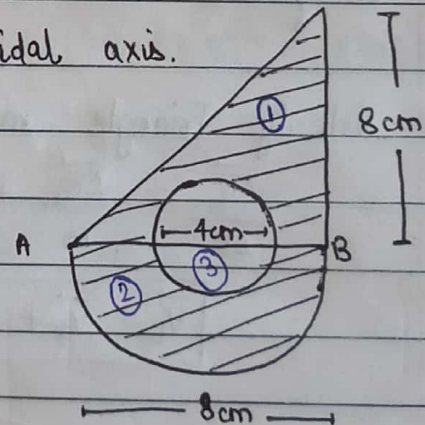
$$= \frac{1}{(2)^2} \left[\frac{2a^4}{3} + a^4 - \frac{2a^4}{5} - \frac{a^4}{2} \right] = \frac{23a^4}{120}$$



Q. Find the M.I. about AB and about centroidal axis.

Solⁿ $\bar{x} = \frac{32 \times \frac{16}{3} + 8\pi \cdot 4 - 4\pi \cdot 4}{32 + 8\pi - 4\pi} = 4.956$

$$\bar{y} = \frac{32 \times \frac{16}{3} + 8\pi \cdot \frac{16}{3} - 0}{32 + 8\pi - 4\pi} = 4.77$$



$$I_{AB} = \frac{1}{12} \cdot 8 \cdot (8)^3 + \frac{1}{8} \pi (4)^4 - \frac{1}{4} \pi (2)^4 = 429.22$$

$$I_{xx} \text{ (about centroidal axis)} = \frac{1}{2} (8)^4 + 32 \times (4.77)^2 + \frac{\pi (4)^4}{8} + 8\pi (4.77)^2 - \frac{\pi (2)^4}{4} - 4\pi \cdot (4.77)^2$$

$$= 429.22 + 1013.74 = 1442.96$$

8/3/18

Page :

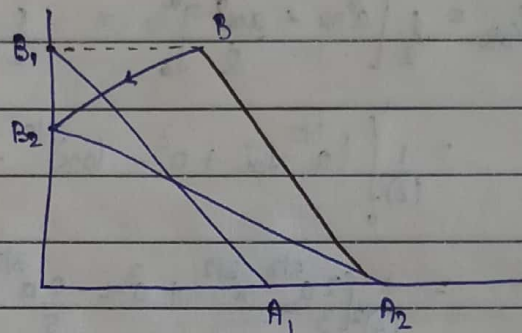
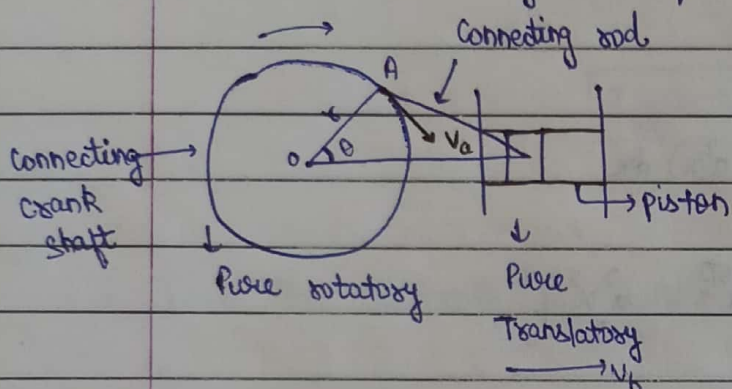
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Kinematics of a Rigid body

General Plane motion:

- Translational
- Rotational

If a body is in both translational and rotational motion, then it is said to be in general plane motion.



General plane motion

1. Triangle method
2. Integrated centre method

(i) Velocity Triangle method:

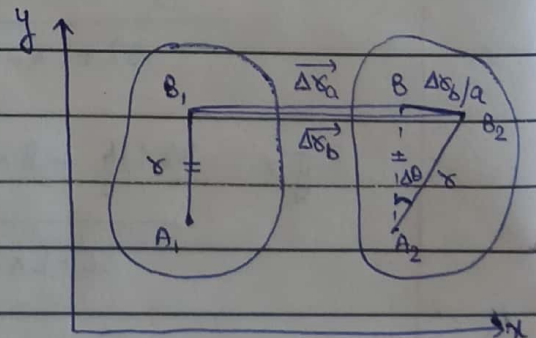
$$\frac{\Delta \vec{r}_B}{\Delta t} = \frac{\Delta \vec{r}_A}{\Delta t} + \frac{\Delta \vec{r}_{B/A}}{\Delta t}$$

$$\vec{V}_B = \vec{V}_A + \vec{V}_{B/A}$$

$$|\vec{V}_{B/A}| = r\omega$$

$$\vec{V}_A = r\omega_{OA}$$

$\vec{V}_{B/A} \rightarrow$ relative velocity due to rotational motion



Q. Find the centroid and moment of inertia.

Soln

$$\bar{y} = \frac{A_1 x_1 - A_2 x_2 - A_3 x_3}{A_1 - A_2 - A_3}$$

$$\bar{y} = \frac{(150)^2 \times 75 - 2500 \times \left(\frac{350}{3}\right) - \frac{\pi (50)^2 \times 75}{2}}{22500 - 2500 - \frac{2500\pi}{2}}$$

$$\bar{y} = 68.5$$

$$\bar{x} = \frac{A_1 y_1 - A_2 y_2 - A_3 y_3}{A_1 - A_2 - A_3}$$

$$\bar{x} = \frac{22500 \times 75 - 2500 \times \frac{50}{3} - \pi \cdot \frac{2500}{2} \times \left(150 - \frac{4 \times 50}{3\pi}\right)}{22500 - 2500 - \frac{2500\pi}{2}}$$

$$\bar{x} = 70.9$$

Centroid = (70.9, 68.5)

$$I_{xx} = I_{xx1} - I_{xx2} - I_{xx3}$$

$$I_{xx1} = I_{c1} + A_1 h_1^2$$

$$= \frac{150 \times 150^3}{12} + 150 \times 150 \times (75 - 68.5)^2 = 43138125$$

$$I_{xx2} = I_{c2} + A_2 h_2^2$$

$$= \frac{1 \times 50 \times (100)^3}{36} + 2500 \times (35.2)^2$$

$$= 4486488.89$$

$$I_{xx3} = I_{c3} + A_3 h_3^2$$

$$= \frac{1 \times \pi (50)^4}{8} + \frac{\pi (50)^2 \times (16.5)^2}{2}$$

$$= 2618956.25$$

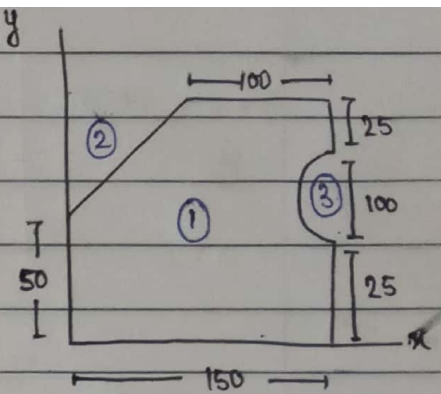
$$I_{xx} = 36032679.86$$

$$I_{yy} = I_{yy1} - I_{yy2} - I_{yy3}$$

$$I_{yy1} = I_{c1} + A_1 h_1^2$$

$$= \frac{150 \times (150)^3}{12} + 150 \times 150 \times (4.1)^2$$

$$= 42565725$$



$$I_{yy2} = I_{CG2} + A_2 h_2^2$$

$$= \frac{100 \times 50^3}{36} + 2500 \times (54.23)^2$$

$$= 7699454.5$$

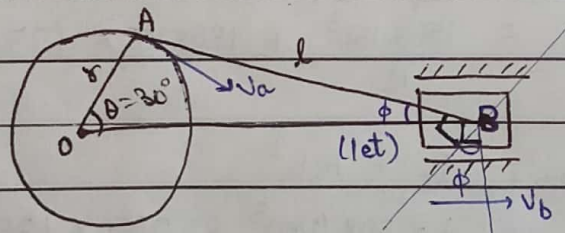
$$I_{yy3} = I_{CG3} + A_3 h_3^2$$

$$= \frac{1}{8} \times (150)^4 + \frac{\pi (150)^2 \times (179.1)^2}{2}$$

$$= 2701104.25$$

$$I_{yy} = 7855166.25$$

- Q. A reciprocating engine mechanism is shown in the figure. The crank OA has a constant angular velocity, $N = 300$ rpm. The crank OA is of length $r = OA = 12$ cm and the connecting rod AB is of 60 cm. Find (i) angular velocity of connecting rod (ii) velocity of the piston when the crank makes an angle of 30° with the horizontal.



Solⁿ (i)

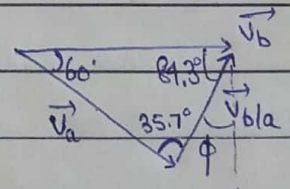
$$v_a = r_{OA} \omega_{OA}$$

$$v_a = 0.12 \times \frac{2\pi N}{60} = 0.12 \times \frac{2\pi \times 300}{60} = 3.77 \text{ m/s}$$

(ii)

$$\vec{v}_b = \vec{v}_a + \vec{v}_{b/a}$$

$$\vec{v}_{b/a} = \vec{v}_b - \vec{v}_a$$



$$\text{In } \triangle OAB, \frac{12}{\sin \theta} = \frac{60}{1/2} \Rightarrow \phi = 5.7^\circ$$

$$\text{By sine rule, } \frac{v_b}{\sin 35.7^\circ} = \frac{3.77}{\sin 84.3^\circ} = \frac{v_{b/a}}{\sin 36^\circ}$$

$$v_{b/a} = 3.28 \text{ m/s} ; v_b = 2.21 \text{ m/s}$$

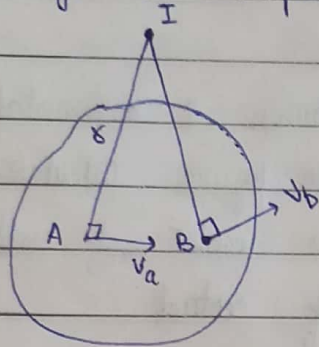
$$v_{b/a} = l \times \omega_{ab}$$

$$\omega_{ab} = \frac{v_{b/a}}{0.60} = \frac{3.28}{0.6} = 5.46 \text{ m/s} \times \text{m}$$

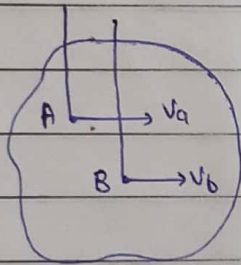
Q. A cylinder of radius 1m

∴ Instantaneous Centre:

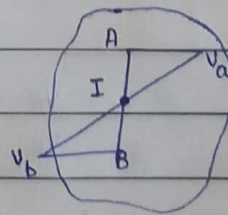
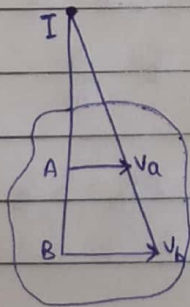
It is the centre about which the body rotates at a particular instant.
It is always in the plane of motion.



$$v_a = \omega \times I_A$$



Since the instantaneous centre lies at ∞ , the body is in translatory motion only.

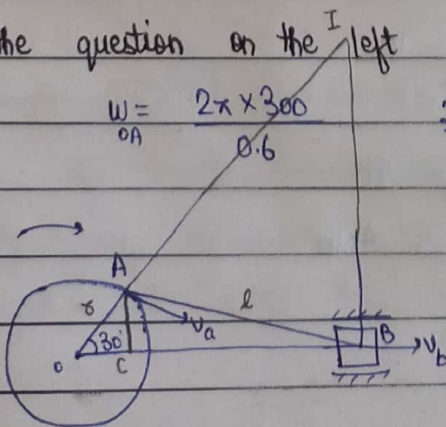


Q. Solve the question on the left page using instantaneous centre method.

Sol.

$$\omega = \frac{2\pi \times 300}{0.6}$$

$$v_a = 0.12 \times \omega = 3.77 \text{ m/s}$$



$$\text{also, } v_a = \omega_{ab} \times I_A$$

$$v_b = \omega_{ab} \times I_B$$

$$v_b = \frac{v_a I_A}{I_B}$$

$$OB = OC + BC = r \cos 30^\circ + l \cos \phi$$

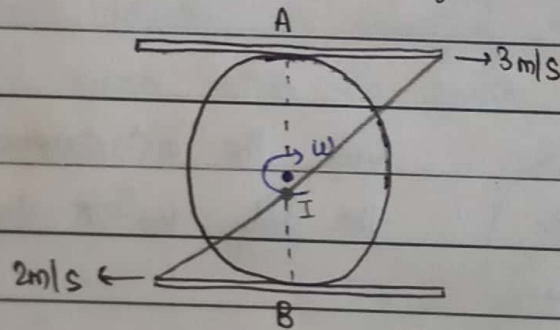
$$I_B = OB \tan 30^\circ = 0.404$$

$$I_{OA} = \frac{I_B}{\sin 30^\circ} \Rightarrow I_A = IO - \text{radius} = 0.689$$

$$So, v_b = v_a \frac{I_B}{I_A} = 2.21 \text{ m/s}$$

$$\omega_{ab} = \frac{v_b}{I_B} = \frac{2.21}{0.404} = 5.5 \text{ rad/sec}$$

- B. A roller of radius 5 cm rides between 2 horizontal bases moving in opposite direction as shown in the figure. Calculate the distance d defining the position of instantaneous centre of roller's position. Assume no slip at the point of contact.



Solⁿ

$$v_a = \omega \times I_A$$

$$I_A = \frac{v_a}{\omega}$$

$$v_b = \omega \times I_B$$

$$I_B = \frac{v_b}{\omega}$$

$$\frac{v_A}{v_b} = \frac{I_A}{I_B} = \frac{3}{2} \Rightarrow 2I_A = 3I_B$$

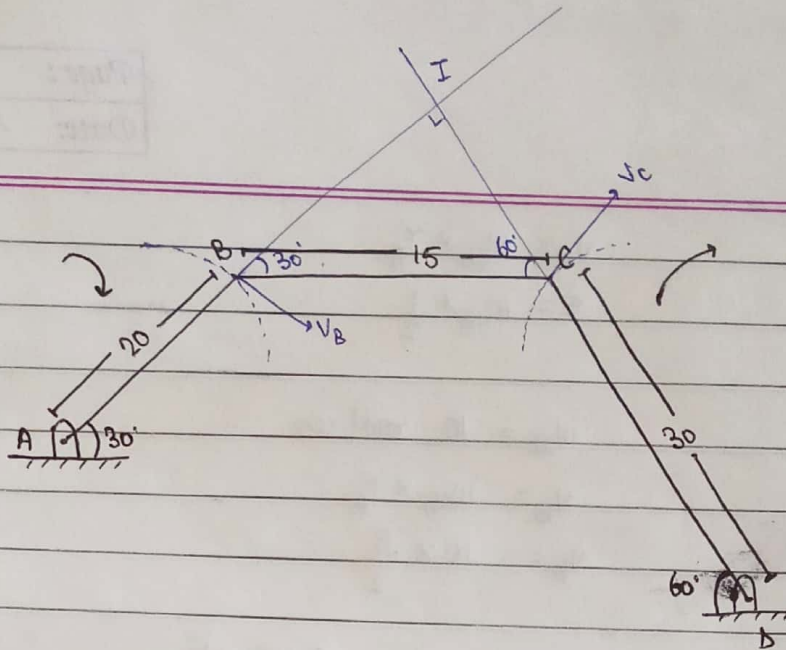
$$\text{and } I_A + I_B = 10 \text{ cm}$$

$$\frac{3}{2} I_B + I_B = 10$$

$$5I_B = 20$$

$$I_B = 4 \text{ cm}$$

Q.



In a 4 bar mechanism ABCD, bar AB rotates clockwise with an angular velocity 5 rad/sec. Find the angular velocities of bar BC and CD when the bar AB makes 30° with the horizontal.

Soln

$$\omega_{AB} = 5 \text{ rad/sec}$$

$$V_B = 0.2 \times 5 = 1 \text{ m/s}$$

$$V_B = \omega_{BC} \times I_B$$

$$I_B = \frac{0.15\sqrt{3}}{2} \text{ m}$$

$$\omega_{BC} = \frac{V_B}{I_B} = \frac{1}{0.15\sqrt{3}} \times 2$$

$$\omega_{BC} = 7.75 \text{ rad/sec}$$

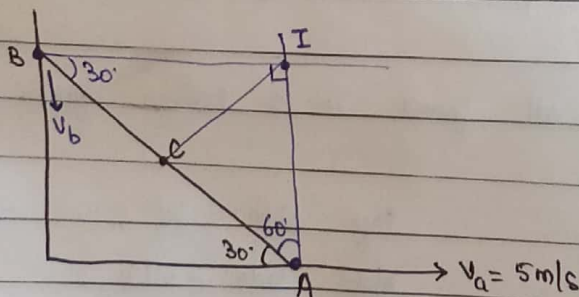
$$V_C = \omega_{BC} \times I_C = 0.58 \text{ m/s}$$

$$I_C = 0.075 \text{ m}$$

$$V_C = \omega_{CD} \times 0.3$$

$$\frac{0.58}{0.3} = \omega_{CD} \Rightarrow \omega_{CD} = 1.98 \text{ rad/sec}$$

Q.



A base of length 1m has its ends A and B constrained to move horizontally and vertically as shown in the figure.

The end A moves with a constant

velocity of 5m/s horizontally. Find (i) ω_{AB} (ii) V_C (iii) velocity of midpoint C of the base at the instant it makes an angle 30° with the horizontal.

(iii)

Sol.ⁿ (i)

$$V_a = \omega_{ab} \times I_a$$

$$5 = \omega_{ab} \times \frac{1}{2}$$

$$\omega_{ab} = 10 \text{ rad/sec}$$

(ii)

$$V_b = \omega_{ab} \times I_b$$

$$V_b = 10 \times \frac{\sqrt{3}}{2}$$

$$V_b = 5\sqrt{3} = 8.65 \text{ m/s}$$

(iii)

$$V_c = \omega_{ab} \times I_c$$

$$I_a = \frac{1}{2}$$

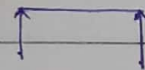
$$I_c = 0.5$$

($\because \Delta ICA$ is equilateral Δ)

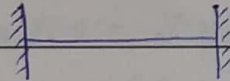
$$V_c = 5 \text{ m/s}$$

Shear force and Bending moment

Simply supported beam - only 2 supports



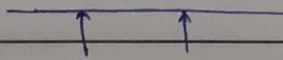
Fixed beam -



Cantilever beam -



Overhanging beam -



Shear force is sum of all forces in a section from one side.

Sign Convention: (left)

$\uparrow (+)$

$\downarrow (-)$

$$R_A = \frac{w}{2}$$

$$R_B = \frac{w}{2}$$

$\curvearrowright (+)$

$\curvearrowleft (-)$

left side: $w/2 - w = -w/2$

right side: $-w/2$

Sign Convention: (right)

$\uparrow (-)$

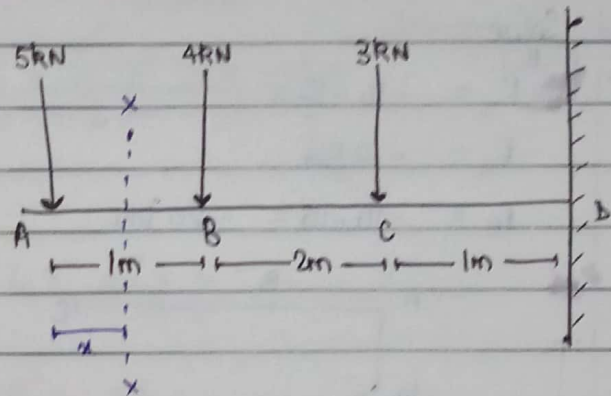
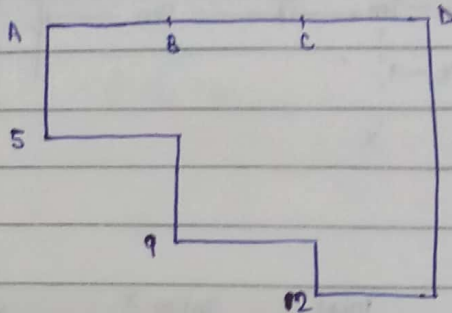
$\downarrow (+)$

$\curvearrowright (-)$

$\curvearrowleft (+)$

Q. Find the shear force and bending moment.

Soln



Shear Force diagram

A-B:

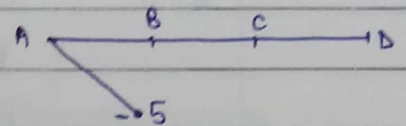
$$M_x = -5x$$

(about XX)

for, $x = 0 \text{ to } 1$

$$M_A = 0$$

$$M_B = -5 \text{ kNm}$$



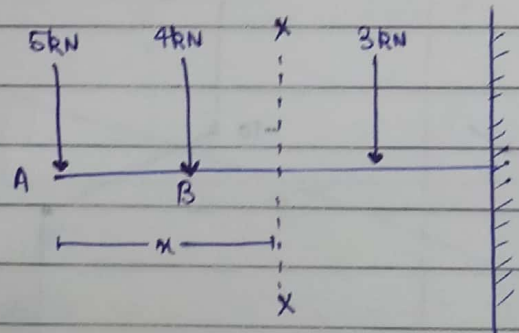
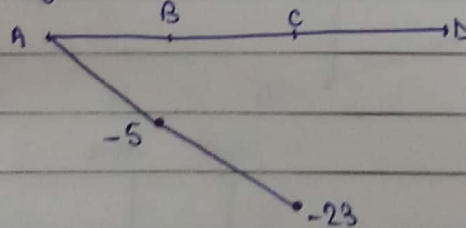
B-C:

$$M_x = -5x - 4(x-1)$$

for, $x = 1 \text{ to } 3$

$$M_B = -5 \text{ kNm}$$

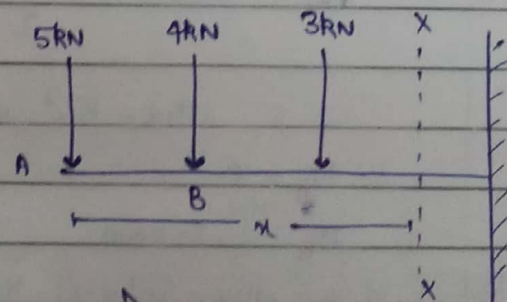
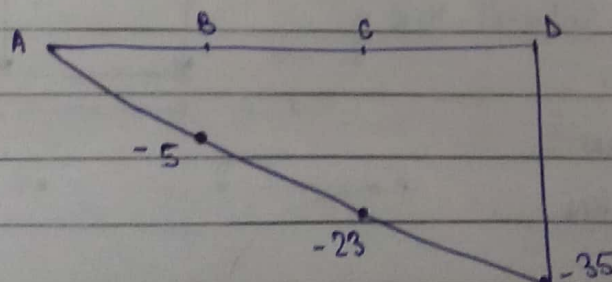
$$M_C = -23 \text{ kNm}$$



C-D: $M_x = -5x - 4(x-1) - 3(x-3)$

for, $x = 3 \text{ to } 4$

$$M_D = -35 \text{ kNm}$$



Q. Solve:

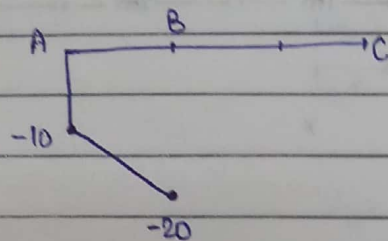
Sol.

A-B:

$$F_x = -10 - 10x$$

$$F_A = -10 \text{ RN}$$

$$F_B = -10 - 10 = -20 \text{ RN}$$

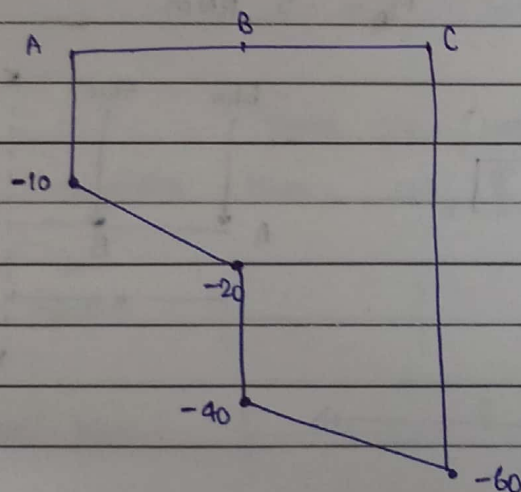


B-C:

$$F_x = -30 - 10x$$

$$F_B = -40 \text{ RN}$$

$$F_C = -30 - 30 = -60 \text{ RN}$$



Shear force diagram

Bending force:

#

$$M_x = -10x - 10x \cdot \frac{x}{2}$$

$$= -10x - 5x^2 \rightarrow \text{parabolic}$$

$$M_A = 0$$

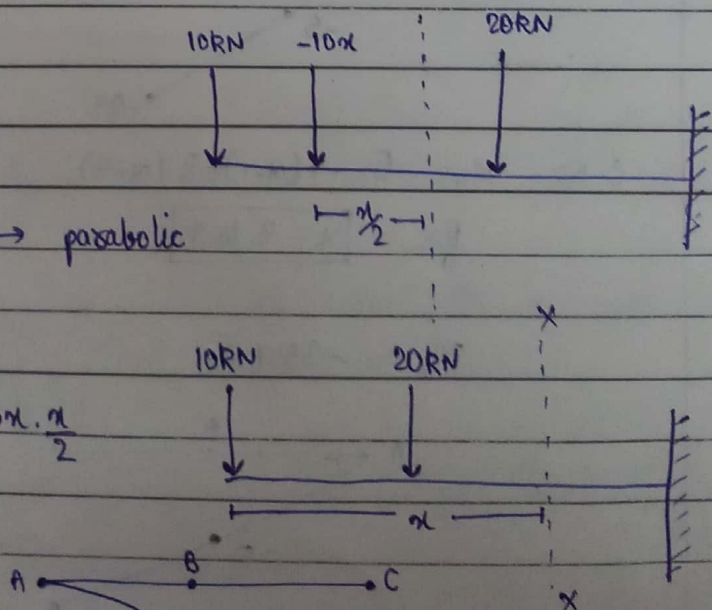
$$M_B = -15 \text{ kNm}$$

#

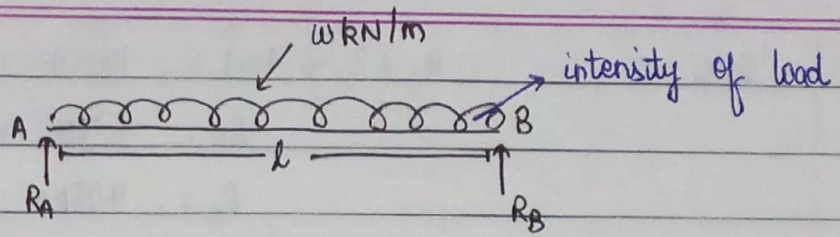
$$M_x = -10x - 20(x-1) - 10x \cdot \frac{x}{2}$$

$$M_B = -15 \text{ kN}$$

$$M_C = -115 \text{ kN}$$



Q. Solve:



Solⁿ

$$\sum M_B = 0 ;$$

$$R_A \times l = w l \times \frac{l}{2}$$

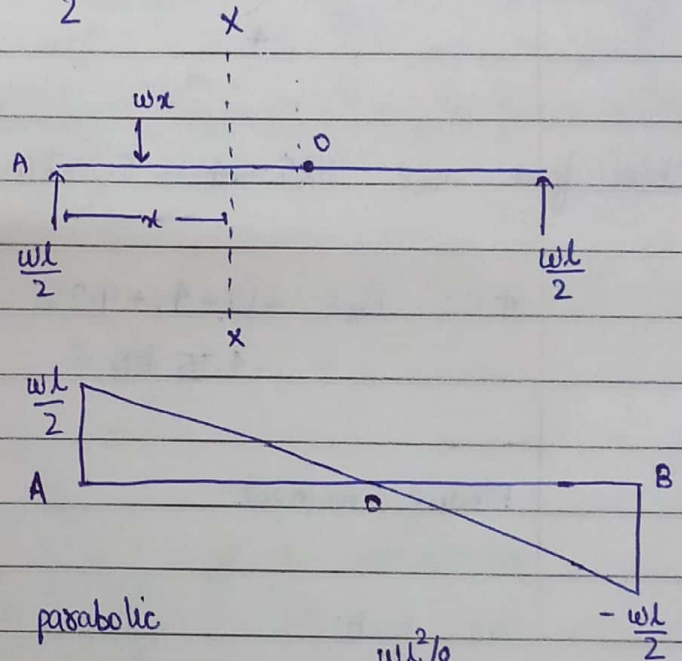
$$R_A = \frac{w l}{2}$$

$$\text{Hky, } R_B = \frac{w l}{2}$$

$$F_x = \frac{w l}{2} - w x$$

$$F_A = \frac{w l}{2} ; F_B = \frac{w l}{2} - w l = -\frac{w l}{2}$$

$$F_{\text{mid-pt.}} = \frac{w l}{2} - \frac{w l}{2} = 0$$



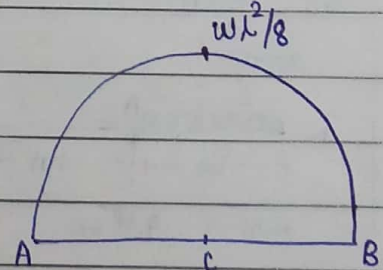
Bending force:

$$M_x = \frac{w l \cdot x}{2} - \frac{w x \cdot x}{2} \rightarrow \text{parabolic}$$

$$x=0, M_A = 0$$

$$x=l, M_B = 0$$

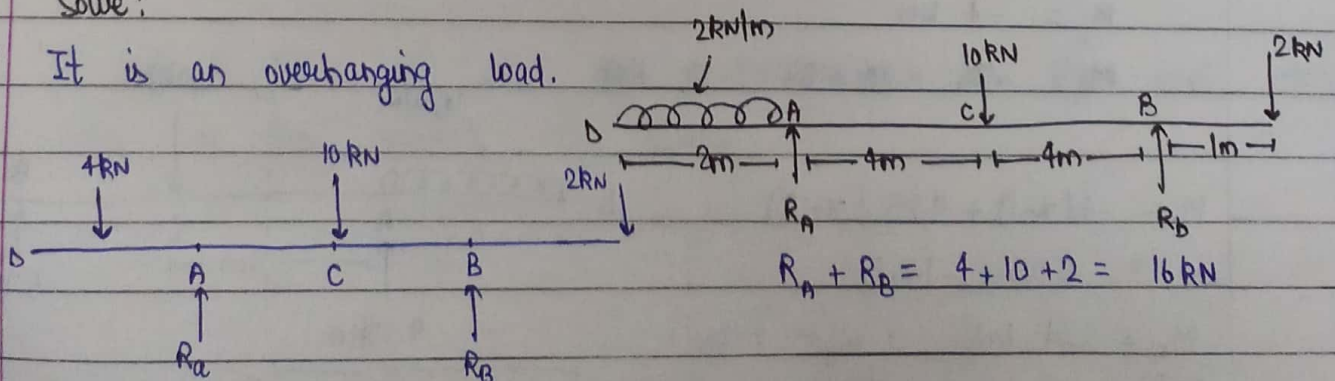
$$x = \frac{l}{2}, M_C = \frac{w l^2}{8}$$



Q. Solve:

Solⁿ

It is an overhanging load.

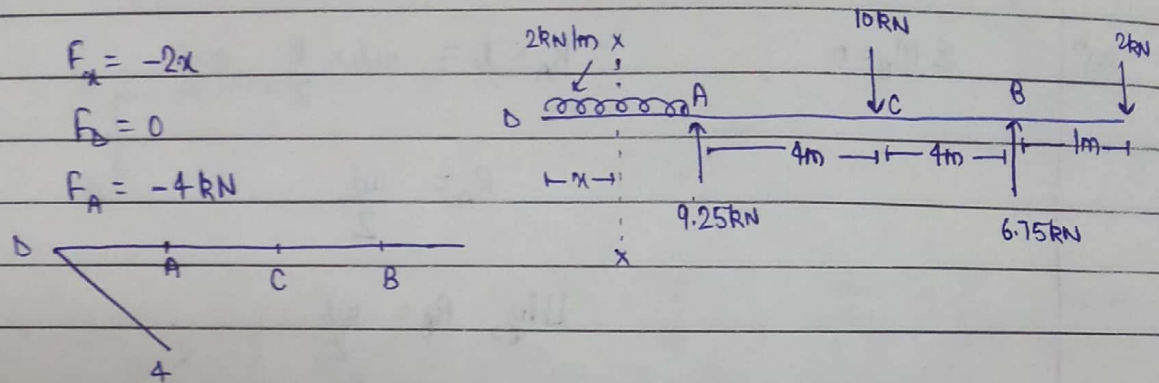


$$R_A + R_B = 4 + 10 + 2 = 16 \text{ kN}$$

$$\sum M_A = 0; \quad R_B \times 8 + 4 \times 1 = 10 \times 4 + 2 \times 9$$

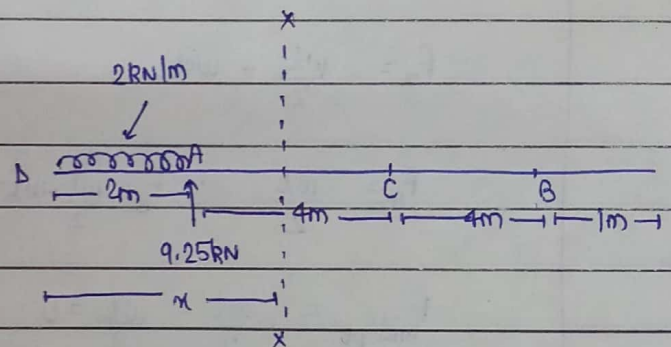
$$R_B = 6.75 \text{ kN}$$

$$R_A = 9.25 \text{ kN}$$



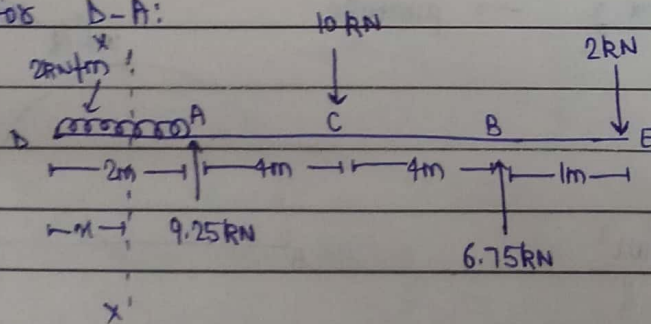
Shear force: $F_x = 9.25 - 4 = 5.25 \text{ kN}$

at C: $F_x = -10 - 4 + 9.25$
 $= -4.75 \text{ kN}$



Bending moment:

For D-A:



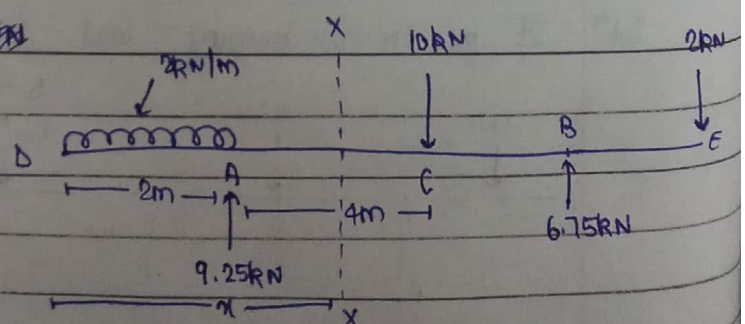
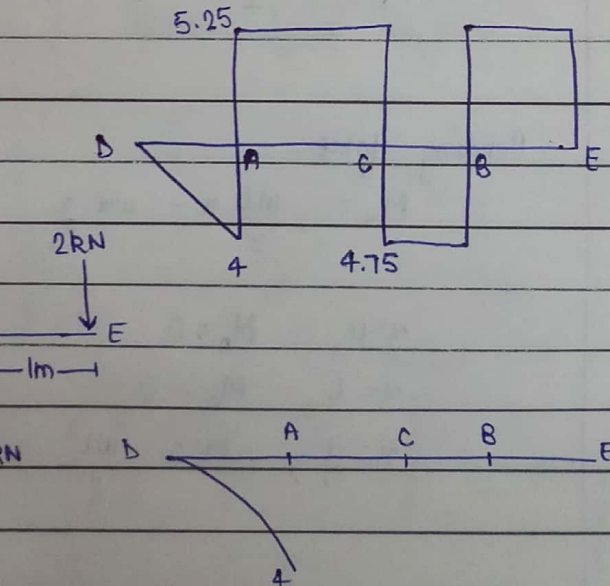
$$M_x = -2x \cdot \frac{x}{2} = -x^2$$

$$M_2 = -4 \text{ kN}$$

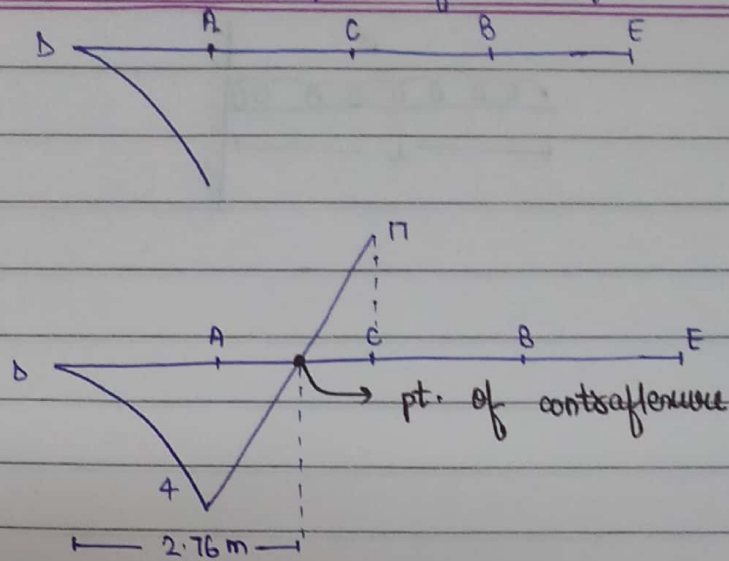
$$M_x = -4(x-1) + 9.25(x-2)$$

$$= 5.25x - 14.5$$

$$M_A = -4 \text{ kN} \quad (x=2) \quad ; \quad M_C = 17 \text{ kN} \quad (x=6)$$

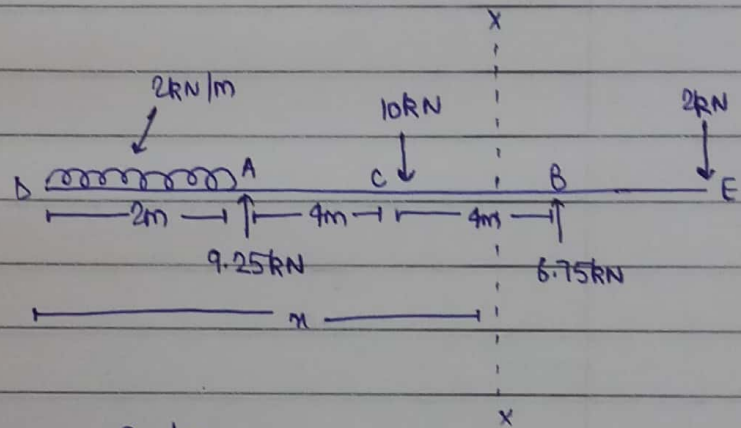


The point on a beam where the bending moment is zero is called point of contraflexure.



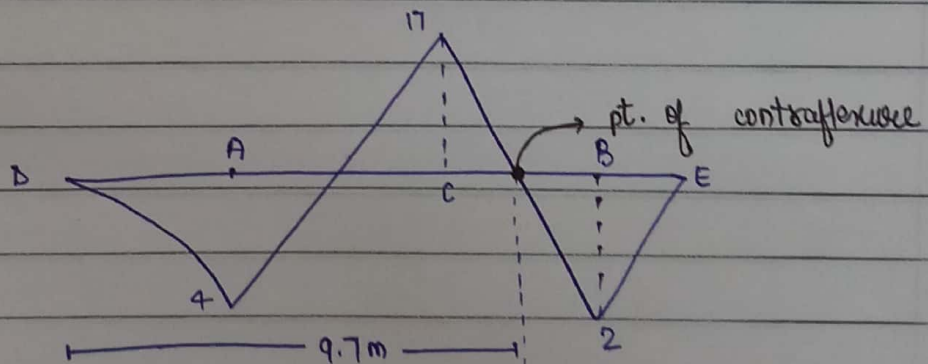
for C-B:

$$M_x = 9.25(x-2) - 4(x-1) - 10(x-6)$$



$$M_C = 17 \text{ kN} \quad (x=6)$$

$$M_B = -2 \text{ kN} \quad (x=10)$$



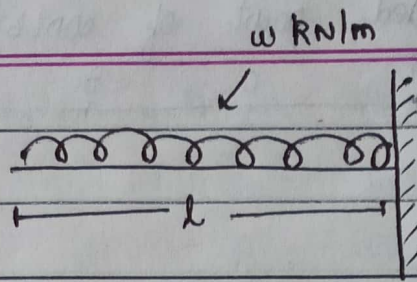
for B-E:

$$M_x = 6.75(x-10) - 4(x-1)$$

$$M_B = -2 \text{ kN} ; M_E = 0$$

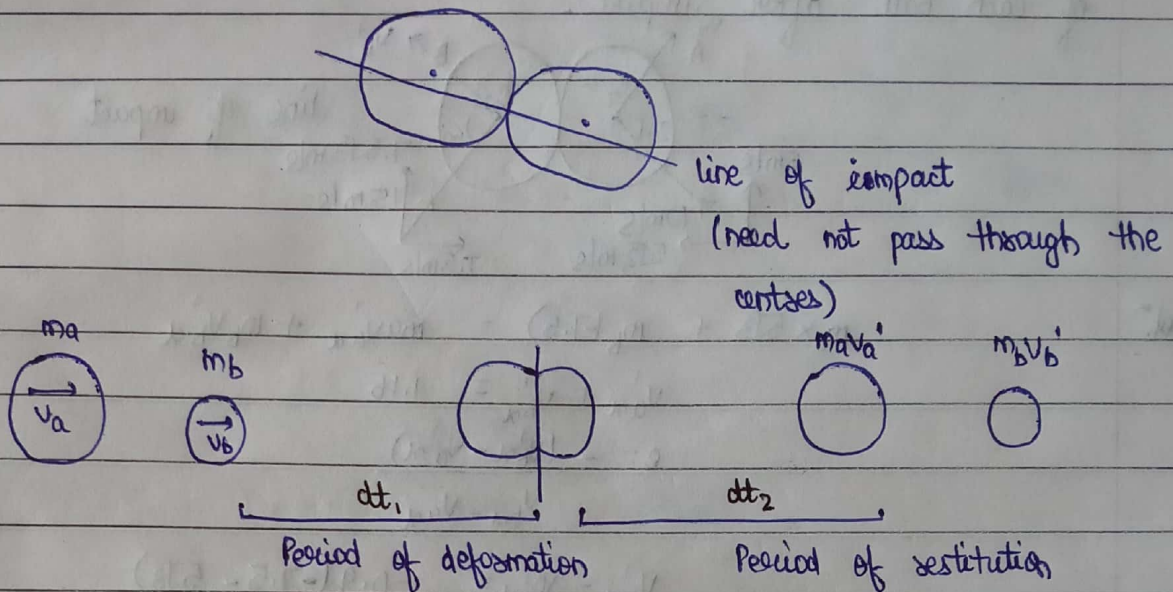
D and E are not point of contraflexures as the point always lies in the middle of beam.

Q. Solve :



Soln

Collision or Impact



Impulse:

$$I \Rightarrow F \times t = m(v - u)$$

$$e = \frac{\text{impulse during restitution}}{\text{impulse during deformation}}$$

$$\text{for body A, } e = \frac{m_A(v - v_A')}{m_A(v_A - v)} = \frac{a}{b}$$

$$\text{for body B, } e = \frac{m_B(v - v_B')}{m_B(v_B - v)} = \frac{c}{d}$$

$$\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$$

$$e = \frac{-(v_B' - v_A')}{v_B - v_A} = -\frac{(\text{velocity of separation})}{(\text{velocity of approach})}$$

for perfectly elastic, $e = 1$, $v_A + v_A' = v_B + v_B'$

for $e < 1$, there is always some loss in energy.

$$\text{Loss of KE} = \frac{2m_A m_B}{m_A + m_B} (v_A - v_B)^2 (1 - e^2)$$

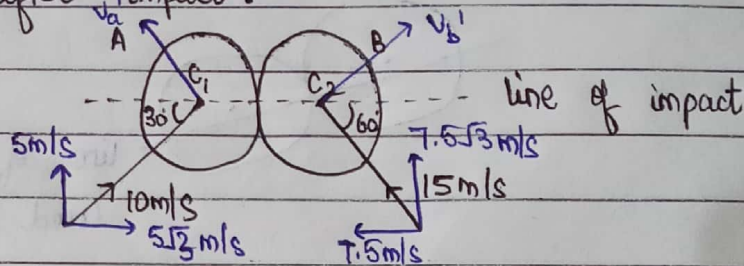
- All transfer of energy takes place along the line of impact.

Apply conservation of momentum along the line of impact.

Page:

Date: / /

- Q. 2 identical frictionless balls strike each other. If the coefficient of restitution is 0.9, determine the magnitude and direction of velocity of each ball after impact.



Sol.

$$m_A \times 5\sqrt{3} + m_B (-7.5) = m_A v_{Ax}' + m_B v_{Bx}'$$

$$v_{Ax}' + v_{Bx}' = 1.16$$

$$e = -\frac{(v_{Bx}' - v_{Ax}')}{v_{Bx} - v_{Ax}}$$

$$v_{Bx} - v_{Ax}$$

$$v_{Bx}' - v_{Ax}' = -0.9(-7.5 - 5\sqrt{3})$$

$$v_{Bx}' - v_{Ax}' = 14.535$$

$$v_{Bx}' + v_{Ax}' = 1.16$$

$$2v_{Bx}' = 15.695$$

$$v_{Bx}' = 7.8475 \text{ m/s}$$

$$v_{Ax}' = -6.68 \text{ m/s}$$

$$v_{Ay} = v_{Ay}' \text{ and } v_{By} = v_{By}'$$

$$v_A' = \sqrt{v_{Ax}'^2 + v_{Ay}'^2}$$

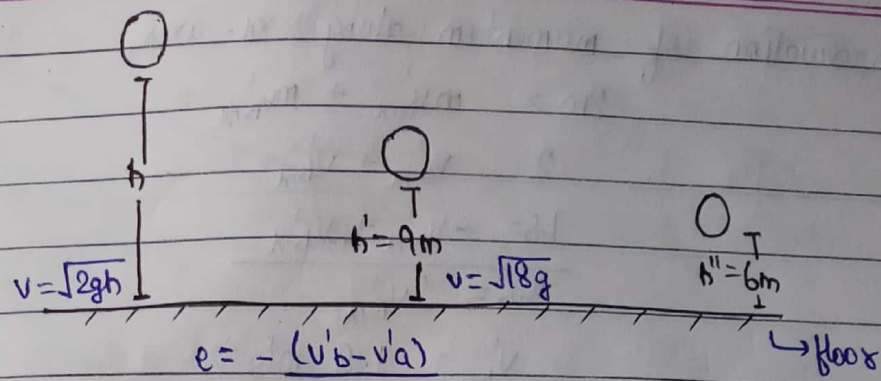
$$v_A' = 8.3 \text{ m/s}$$

$$v_B' = \sqrt{v_{Bx}'^2 + v_{By}'^2}$$

$$v_B' = 15.18 \text{ m/s}$$

$$\theta = \tan^{-1}\left(\frac{5}{6.68}\right)$$

- Q. A glass ball is dropped onto a smooth horizontal floor from which it bounces to a height of 9 m. On the second bounce, it rises to a height of 6 m. From what height, the ball was dropped and what is the coeff. of restitution?



Solⁿ

$$e = -\frac{v'_b - v'_a}{v_b - v_a}$$

$$e = \frac{v'_a}{v_a}$$

$$v^2 - u^2 = 2gh$$

$$e^2(2gh) = 2(+6)g$$

$$e^2h = 9$$

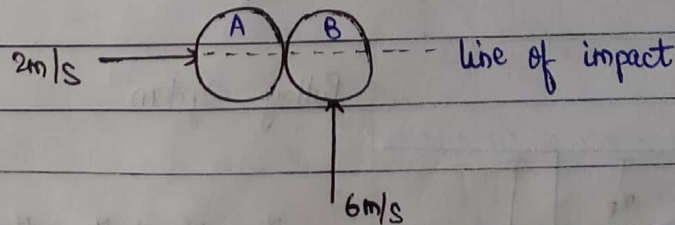
$$e^2(18g) = 2g(6)^2$$

$$3e^2 = 2$$

$$e = \sqrt{\frac{2}{3}}$$

$$\Rightarrow h = \frac{9}{e^2} = \frac{9 \times 3}{2} = 13.5\text{ m}$$

Q. 2 identical balls strike each other as shown in the figure. If $e = 0.8$, determine the velocities after the impact.



Solⁿ

$$e = -\frac{v'_b - v'_a}{v_b - v_a} = 0.8$$

$$\frac{-(v'_{bx} - v'_{ax})}{v_{bx} - v_{ax}} = 0.8$$

$$\frac{-(v'_{bx} - v'_{ax})}{-2} = 0.8 \Rightarrow v'_{bx} - v'_{ax} = 1.6$$

Conservation of momentum along x-axis,

$$2m = mv'_{ax} + mv'_{bx}$$

$$2 = v'_{ax} + v'_{bx}$$

$$1.6 = -v'_{ax} + v'_{bx}$$

$$3.6 = 2v'_{bx}$$

$$v'_{bx} = 1.8 \text{ m/s}$$

$$v'_{ax} = 0.2 \text{ m/s}$$

$$e = -\frac{(v'_{by} - v'_{ay})}{v_{by} - v_{ay}} = 0.8$$

$$-(v'_{by} - v'_{ay}) = 0.8 \times 6$$

$$v'_{ay} - v'_{by} = 4.8$$

Conservation of momentum along y-axis,

$$6m = mv'_{ay} + mv'_{by}$$

$$6 = v'_{ay} + v'_{by}$$

$$4.8 = v'_{ay} - v'_{by}$$

$$10.8 = 2v'_{ay}$$

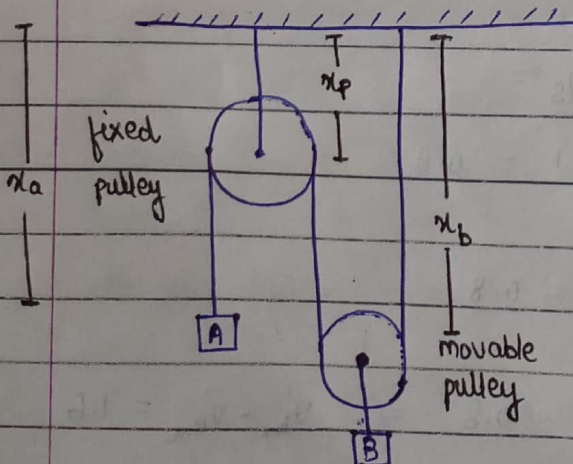
$$v'_{ay} = 5.4 \text{ m/s}$$

$$v'_{by} = 0.6 \text{ m/s}$$

$$\vec{v}'_a = 0.2\hat{i} + 5.4\hat{j}$$

$$\vec{v}'_b = 1.8\hat{i} + 0.6\hat{j}$$

Pulley System

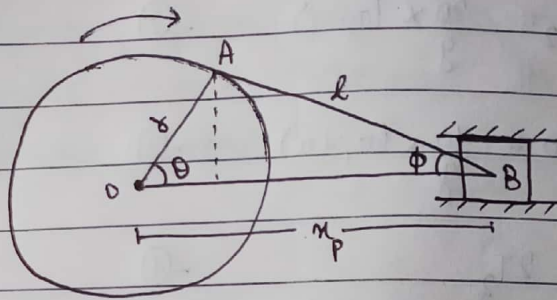


$$x_a - x_p + x_b - x_p + x_b = l$$

$$x_a + 2x_b = l + 2x_p$$

$$acc._a + 2acc._b = 0$$

$$acc._b = -\frac{acc._a}{2}$$



$$\frac{r}{\sin \phi} = \frac{l}{\sin \theta}$$

$$\sin \phi = \frac{r \sin \theta}{l}$$

$$x_p = r \cos \theta + l \cos \phi$$

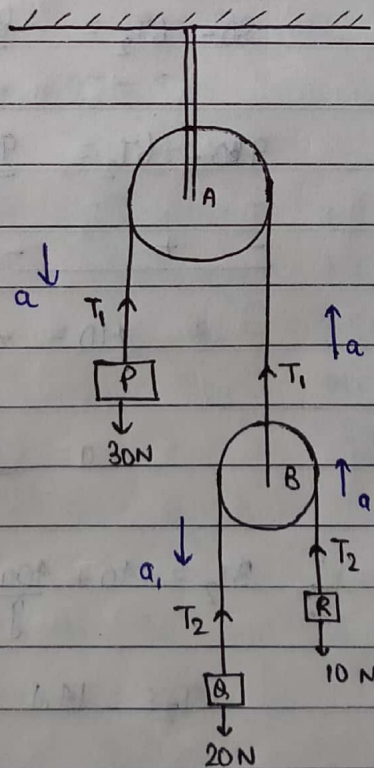
$$x_p = r \cos \omega t + l \sqrt{1 - \frac{r^2 \sin^2 \omega t}{l^2}} \quad [\because \theta = \omega t]$$

$$x_p = r \cos \omega t + l \quad (\text{approx.})$$

$$v_p = -r \omega \sin \omega t$$

$$a_{c.p} = -r \omega^2 \cos \omega t$$

- Q. A system of weights connected by spring and passing over 2 pulleys A and B are arranged as shown. Find the acc. of each weights. Neglect friction and inertia of pulleys.



Sol.

Absolute acceleration of R = $a_1 + a$

Absolute acceleration of Q = $a_1 - a$

$$\text{for P, } 30 - T_1 = \frac{30 \times a}{g} \quad \text{--- (1)}$$

$$\text{for } Q, \quad 20 - T_2 = \frac{20 \times (a_1 - a)}{g} \quad - (2)$$

$$\text{for } R, \quad T_2 - 10 = \frac{10 \times (a_1 + a)}{g} \quad - (3)$$

$$T_1 = 2T_2 \quad - (4)$$

$$\text{So, } 30 - 2T_2 = \frac{30 \times a}{g}$$

from eq. (2) and (3),

$$20 - T_2 = \frac{20a_1}{g} - \frac{20a}{g}$$

$$-20 + 2T_2 = \frac{20a_1}{g} + \frac{20a}{g}$$

$$\begin{array}{r} + \quad - \quad - \quad - \\ \hline 40 - 3T_2 = -\frac{40a}{g} \end{array}$$

$$80 - 6T_2 = -\frac{80a}{g}$$

$$90 - 6T_2 = \frac{90a}{g}$$

$$\begin{array}{r} - \quad + \quad - \\ \hline \end{array}$$

$$+10 = +\frac{170a}{g}$$

$$a = \frac{g}{17}$$

$$3T_2 = 40 + \frac{40a}{g} = 40 + \frac{40 \times g}{g \times 17} = 42.3$$

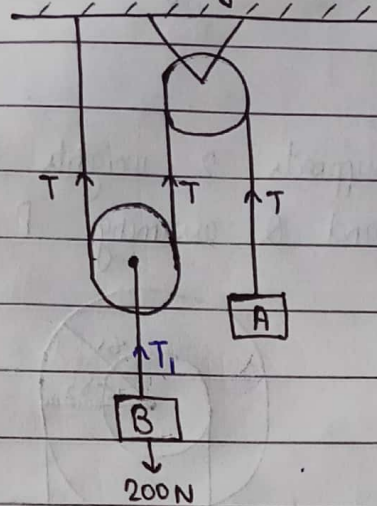
$$T_2 = 14.1 \text{ N} ; \quad T_1 = 28.2 \text{ N}$$

$$4.1 = \frac{10}{g} \left(a_1 + \frac{g}{17} \right)$$

$$\frac{4.1g}{10} - \frac{g}{17} = a_1 = 0.352g$$

Q. A system of connected bodies has frictionless pulleys and of negligible weight. Determine the value of weight of A so that B moves upward with $0.6g$ (ii) moves downward with $0.6g$.

Solⁿ



(ii)

$$acc_A = 2acc_B \quad \text{--- (1)}$$

$$\text{for B, } 200 - 2T_1 = \frac{200 \times (0.6g)}{g}$$

$$200 - 2T_1 = 120$$

$$2T_1 = 80 \Rightarrow T_1 = 40 \text{ N}$$

$$2T = T_1$$

$$T = 20 \text{ N}$$

$$\text{for A, } T - mg = m(1.2g)$$

$$20 - mg = 1.2mg$$

$$20 = 2.2mg$$

$$m = \frac{20}{2.2g} = 1.81 \text{ kg}$$

$$\text{wt. of A} = 18.18 \text{ N}$$

(i) Mass comes out to be -ve (of A).

$$acc_A = 2acc_B$$

$$\text{for B, } T_1 - 200 = \frac{200(0.6g)}{g}$$

$$T_1 - 200 = 120$$

$$T_1 = 320 \text{ N}$$

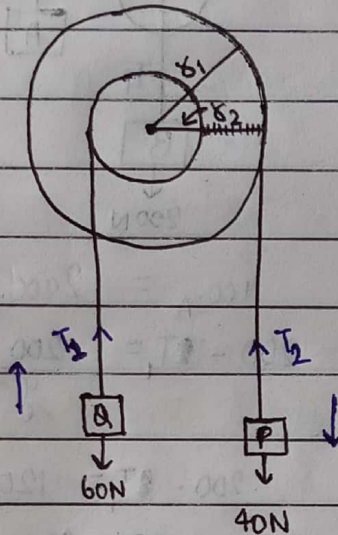
$$2T = T_1 \Rightarrow T = 160 \text{ N}$$

$$\text{for A, } mg - T = m(1.2g)$$

$$160 = -0.2mg$$

$$mg = \frac{-1600}{2} = -800 \text{ N}$$

- Q. A 2-step pulley supports 2 weights P and Q. Find the downward acceleration of P and Q assuming P is of 40 N and Q is of 60 N.
Given $r_1 = 2r_2$



Sol. P would move downward as its moment is greater ($M_p = 40r_1$).

$$M_q = 60r_2$$

$$\text{for P, } 40 - T_2 = \frac{40}{g} (a_p)$$

$$40 - T_2 = \frac{40}{g} \times a_p$$

$$\text{for Q, } T_1 - 60 = \frac{60}{g} \times a_q$$

We know, $a = r\alpha$
 $\rightarrow$ constant

$$\text{So, } \frac{a_p}{r_1} = \frac{a_q}{r_2}$$

$$a_p = 2a_q$$

$$-T_2 \times a_p + T_1 \times a_q = 0$$

$$T_1 \times a_q = T_2 \times 2a_q$$

$$T_1 = 2T_2$$

$$(40 - T_2 = \frac{40 \times 2a_a}{g}) \times 2$$

$$-60 + 2T_2 = \frac{60 \times a_a}{g}$$

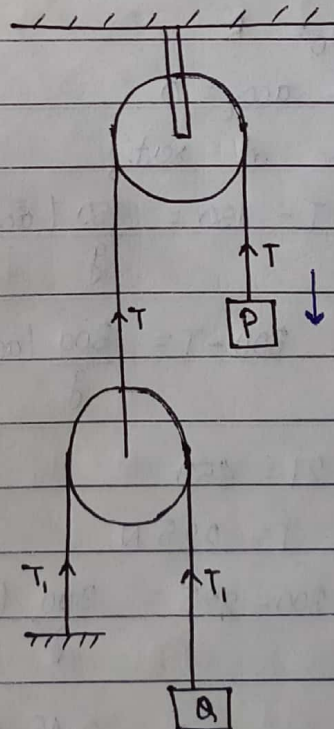
$$80 - 2T_2 = \frac{80 \times 2 \times a_a}{g}$$

$$20 = \frac{22a_a}{g}$$

$$a_a = \frac{g}{11}$$

$$a_p = \frac{2g}{11}$$

Q. Neglect inertia of the pulleys, find the acceleration of P if mass of both blocks is equal.



Sol.ⁿ

$$T = 2T_1$$

$$\text{for } P, \quad mg - T = ma_p$$

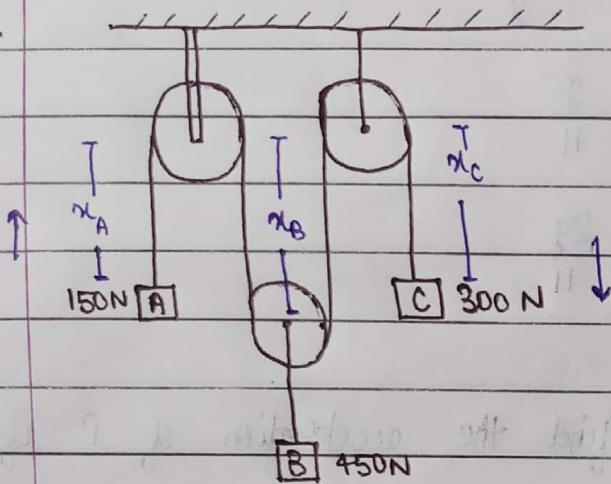
$$\text{for } Q, \quad T_1 - mg = ma_a$$

$$-T a_p + T_1 a_a = 0$$

$$a_a = 2a_p \quad (\because T = 2T_1)$$

$$\begin{aligned}
 mg - 2T_1 &= ma_p \\
 (-mg + T_1 &= 2ma_p) \times 2 \\
 -2mg + 2T_1 &= 4ma_p \\
 \hline
 -mg &= 5ma_p \\
 a_p &= \frac{-g}{5}
 \end{aligned}$$

Q.



Find the acceleration of the blocks A, B and C.

Sol.

$$a_A + 2a_B + a_C = 0$$

$$acc_A + 2acc_B + acc_C = 0$$

Assuming B to be at rest,

$$\text{for A, } T - 150 = \frac{150 (acc_A)}{g}$$

$$\text{for C, } 300 - T = \frac{300 (acc_C)}{g}$$

$$2T = 450$$

$$T = 225 \text{ N}$$

$$300 - 225 = \frac{300 (acc_C)}{g}$$

$$acc_C = 2.45 \text{ m/s}^2$$

$$225 - 150 = \frac{150 (acc_A)}{g}$$

$$\frac{75g}{150} = acc_A$$

$$acc_A = 4.9 \text{ m/s}^2$$

$$2acc_B = -2.45$$

$$acc_B = -1.225 \text{ m/s}^2$$

Curvilinear motion

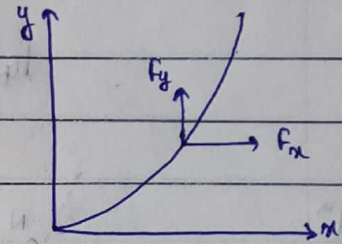
$$F_x = ma_x$$

$$F_y = ma_y$$

$$a_t = \frac{dv}{dt}$$

$$a_n = \frac{v^2}{r}$$

normal acc.



$$F_x - ma_x = 0$$

$$F_x + (-ma_x) = 0 \rightarrow \text{eq. of dynamic equilibrium}$$

$\rightarrow$ D'Alembert's eq.

$-ma_x \rightarrow$ Inertia Force

- Q. A car is travelling on a curved road of radius 20 m at a speed of 72 km/hr. Find the a_t and a_n .

Sol.

$$v = \frac{72 \times 5}{18} = 20 \text{ m/s}$$

$$a_n = \frac{v^2}{r} = \frac{400}{20} = 20 \text{ m/s}^2$$

$$a_t = 0 \text{ m/s}^2$$

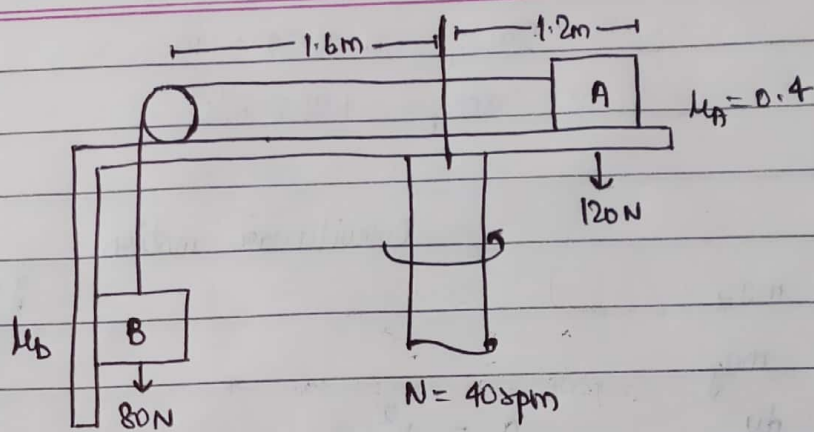
- Q. If brakes are applied to slow down the speed at $v = 10 \text{ m/s}$ in 10 sec., find a_t and a_n .

Sol.

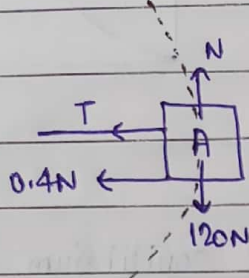
$$a_n = 20 \text{ m/s}^2$$

$$a_t = \frac{10}{10} = 1 \text{ m/s}^2$$

- Q. The frame shown in the figure rotates about a vertical axis. The coeff. of friction under block A is 0.4. Determine the coeff. of friction of block B if B starts to rise when frame rotates at 40 rpm.



Sol.ⁿ



$$a_N = \frac{v^2}{r} = \frac{\omega^2 r}{1} = \omega^2 r$$

$$T + 0.4N = F_N = ma_N$$

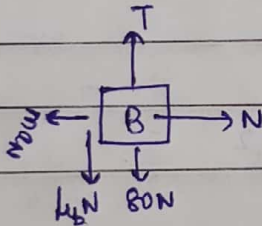
$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 40}{60} = 4.19 \text{ rad/sec.}$$

$$a_N = \omega^2 r = 1.2 \times (4.19)^2$$

$$a_N = 21.06 \text{ m/s}^2$$

$$T = -0.4 \times 120 + 12 \times 21.06$$

$$T = 209.72 \text{ N}$$



$$a_N = 1.6 \times (4.19)^2$$

$$a_N = 28.08 \text{ m/s}^2$$

$$T = \mu_B N + 80$$

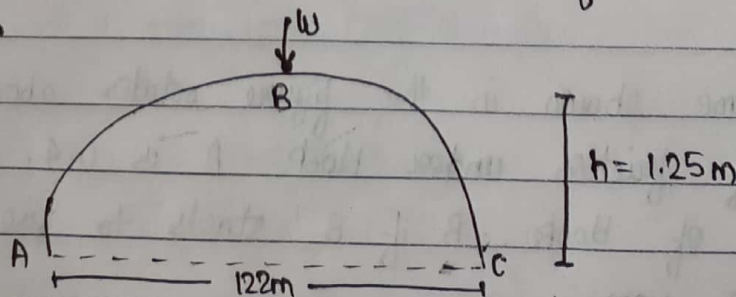
$$\mu_B N = 129.72 \text{ N}$$

$$N = ma_N = 224.64$$

$$\mu_B = \mu_A = 0.57$$

8. An automobile of weight W travels with a speed of $v = 100 \text{ km/h}$ over a vertical curve ABC of parabolic shape. Determine the total pressure exerted by the wheels of the car as it passes point B.

$$n^2 = R_y$$



Sol.ⁿ

$$x^2 = ky$$

$$2x = ky'$$

$$y' = \frac{2x}{k}$$

$$y'' = \frac{2}{k}$$

$$R = \left(\frac{y}{x^2} \right)^{-1} = \left(\frac{-1.25}{(161)^2} \right)^{-1} = -297.68 - 2976.8$$

$$\frac{1}{\rho} = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}} = \frac{2}{297.68}$$

$$\rho = 1468.4 \text{ m}$$

$$W - R = \frac{mv^2}{\rho}$$

$$R = W - \frac{mv^2}{\rho}$$

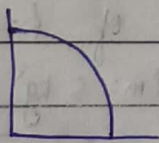
- Q. A car enters a curved road of radius 200 m travelling at a constant speed of 36 km/h. Determine the components of acc. in the x and y dir.ⁿ, 15 seconds after it has entered the road. Also find v_x and v_y .

Sol.ⁿ

$$a_n = \frac{v^2}{r} = \frac{100}{200} = 0.5 \text{ m/s}^2$$

$$a_t = \frac{dv}{dt} = 0$$

$$\omega = \frac{v}{r} = \frac{10}{200} = \frac{1}{20} \text{ rad/sec.}$$



$$x = r \cos \theta = r \cos \omega t$$

$$y = r \sin \theta = r \sin \omega t$$

$$v_x = -r \omega \sin \omega t = -200 \times \frac{1}{20} \sin \left[\frac{1}{20} \times \frac{180}{\pi} \times 15 \right]^\circ = -6.82 \text{ m/s}$$

$$v_y = r \omega \cos \omega t = 200 \times \frac{1}{20} \cos \left[\frac{1}{20} \times \frac{180}{\pi} \times 15 \right]^\circ = 7.71 \text{ m/s}$$

$$a_x = \frac{dv_x}{dt} = -\omega^2 x \cos \omega t = -\left(\frac{1}{20}\right)^2 \times 200 \cos \left[\frac{1}{20} \times \frac{180}{\pi} \times 15\right]^\circ$$

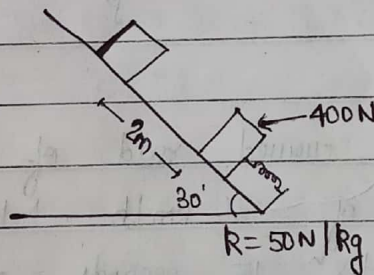
$$a_x = 0.365 \text{ m/s}^2$$

$$a_y = \frac{dv_y}{dt} = -\omega^2 x \sin \omega t = -\left(\frac{1}{20}\right)^2 \times 200 \sin \left[\frac{1}{20} \times \frac{180}{\pi} \times 15\right]^\circ$$

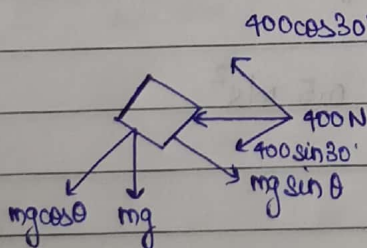
$$a_y = 0.341 \text{ m/s}^2$$

Work and Energy

- Q. A 10 kg block rests on smooth inclined plane. If spring is originally unstretched, determine the total work done by all forces acting on block when a horizontal force 400 N pushes this block up the plane by distance 2m.



Soln



$$\text{Work} = 400 \cos 30^\circ \times 2 - mg \sin \theta \times 2 + \frac{1}{2} kx^2$$

done

$$W = 400 \times \frac{\sqrt{3}}{2} \times 2 - 10 \times 10 \times \frac{1}{2} \times 2 + \frac{1}{2} \times 500 \times 4$$

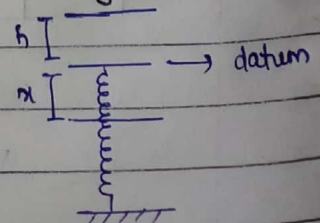
$$W = 400\sqrt{3} - 100 + 100 = 400\sqrt{3} \text{ Nm}$$

- Q. A bomb of mass m is dropped onto a spring of k = 500 N/m from a height 10 m, find max. deflection. (m = 5 kg)

Soln

I case, $mgh = \frac{1}{2} mv^2$

$$v = \sqrt{2gh} = 10.52 \text{ m/s}$$



II case, $\frac{1}{2}mv^2 = \frac{1}{2}Rx^2 - mgx$

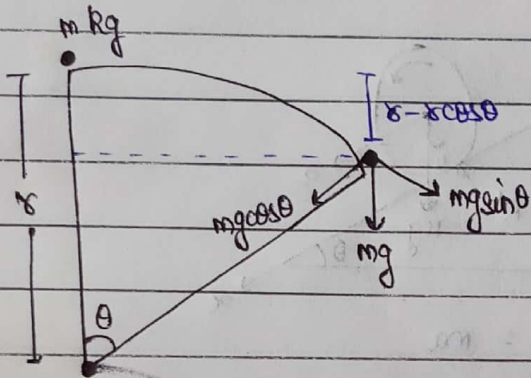
$$500 = 250x^2 - 50x$$

$$5x^2 - x - 10 = 0$$

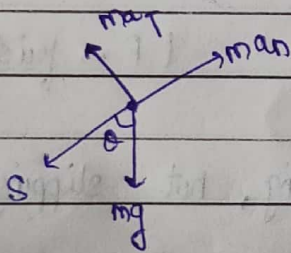
$$x = \frac{1 \pm \sqrt{1+200}}{10}$$

$$x = 26.91 \text{ cm}$$

Find the angle θ at which packet begins to leave the surface.



Solⁿ



$$ma_n - S - mg \cos \theta = 0$$

$$a_n = \frac{v^2}{R}$$

$$S = -mg \cos \theta + \frac{mv^2}{R}$$

When it leaves the surface, $S = 0$

$$\cos \theta = \frac{v^2}{Rg}$$

$$0 = \frac{1}{2}mv^2 - mg(R - R \cos \theta)$$

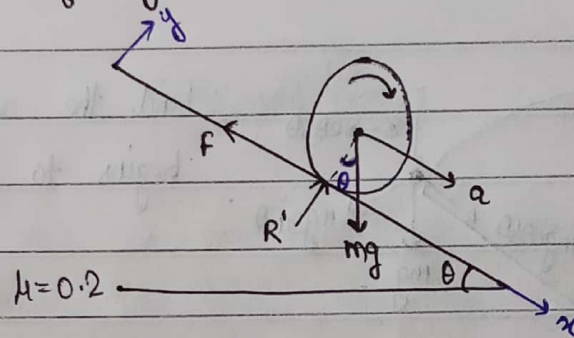
$$\frac{v^2}{2} = gR \left(1 - \frac{v^2}{Rg}\right) = gR(1 - \cos \theta)$$

$$\cos \theta = 2(1 - \cos \theta)$$

$$\therefore \cos \theta = \frac{2}{3}$$

$$\theta = \cos^{-1}\left(\frac{2}{3}\right)$$

- Q. A cylinder of mass m and radius R resting on an inclined plane is released from rest. It rolls down the plane without slipping. Determine (a) acc. of its COM
 (b) max. angle θ of inclined plane for which the body will ~~slip~~ roll without any slip
 (c) max. velocity of COM for $s=1\text{ m}$
 (d) acc. of centre of cylinder under frictionless conditions



Soln (a)

$$mg \sin \theta - F = ma$$

$$R' = mg \cos \theta$$

$$FR = I\alpha$$

$$a = R\alpha$$

($F \rightarrow$ frictional force)

$F \neq \mu R$ as the body is only rolling, not slipping.

$$F = \frac{I\alpha}{R}$$

$$mg \sin \theta - \frac{I\alpha}{R} = ma$$

$$mg \sin \theta - \frac{Ia}{R^2} = ma$$

$$mg \sin \theta = a \left(m + \frac{I}{R^2} \right)$$

$$a = \frac{mg \sin \theta}{m + \frac{I}{R^2}} = \frac{mg \sin \theta}{m + \frac{mR^2}{2R^2}} = \frac{2}{3} g \sin \theta$$

(b)

$$mg \sin \theta - F = \frac{2}{3} mg \sin \theta$$

$$F = \frac{mg \sin \theta}{3}$$

$$F < \mu R$$

$$F < \mu mg \cos \theta$$

$$\frac{mg \sin \theta}{3} < 0.2 mg \cos \theta$$

$$\tan \theta < 0.6$$

$$\theta < \tan^{-1}(0.6)$$

$$\theta < 30.9^\circ$$

(c)

$$v^2 - u^2 = 2as$$

$$v^2 = 2 \times \frac{2}{3} g \sin \theta \times 1$$

$$v^2 = \frac{4}{3} \times 10 \times \sin(30.9^\circ)$$

$$v = 2.56 \text{ m/s}$$

(d)

$$mg \sin \theta - 0 = ma$$

$$a = g \sin \theta$$

It won't roll in this case.

Q. A sphere, a cylinder and a hoop, each having same mass and radius, are released from rest on an inclined plane (θ). If they roll down without slip, compare their acc.

Solⁿ

$$I_{\text{sphere}} = \frac{2}{5} m r^2$$

$$I_{\text{cylinder}} = \frac{1}{2} m r^2$$

$$I_{\text{hoop}} = \frac{1}{2} m r^2$$

$$\text{acc.} = 0.714 g \sin \theta$$

$$> \text{acc.} = 0.66 g \sin \theta$$

$$> \text{acc.} = 0.5 g \sin \theta$$

$$F < \mu R$$

$$F < \mu mg \cos \theta$$

$$\frac{mg \sin \theta}{3} < 0.2 mg \cos \theta$$

$$\tan \theta < 0.6$$

$$\theta < \tan^{-1}(0.6)$$

$$\theta < 30.9^\circ$$

(c)

$$v^2 - u^2 = 2as$$

$$v^2 = \frac{2 \times 2}{3} g \sin \theta \times 1$$

$$v^2 = \frac{4 \times 10 \times \sin(30.9^\circ)}{3}$$

$$v = 2.56 \text{ m/s}$$

(d)

$$mg \sin \theta - 0 = ma$$

$$a = g \sin \theta$$

It won't roll in this case.

Q. A sphere, a cylinder and a hoop, each having same mass and radius, are released from rest on an inclined plane (θ). If they roll down without slip, compare their acc.

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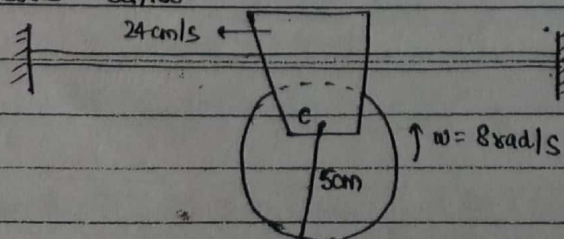
>

$$\text{acc.} = 0.66 g \sin \theta$$

>

$$\text{acc.} = 0.5 g \sin \theta$$

Q. Find the instantaneous centre.

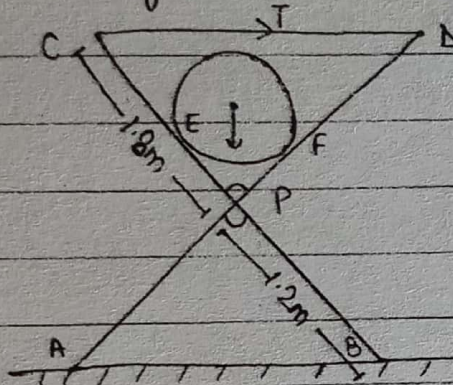
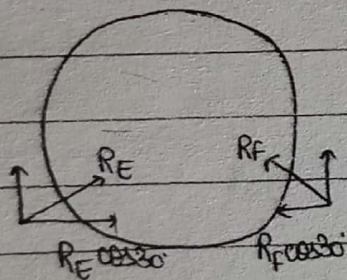


Sol.ⁿ

$$V = 8 \times W$$

$$8 = \frac{24}{8} = 3 \text{ cm}$$

- D. A steel cylinder of 1m diameter and weighing 500N is placed b/w cross beams, making 60° angle and are hinged at P. A rope attached to one near their upper ends prevents them from sliding. Assume flow to be smooth, find tension in rope.

Sol.ⁿ

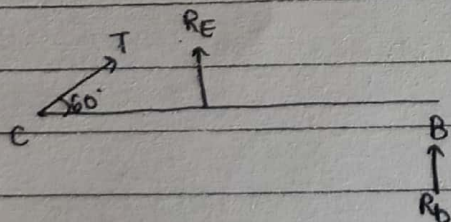
$$R_E \cos 30^\circ = R_F \cos 30^\circ$$

$$R_E = R_F$$

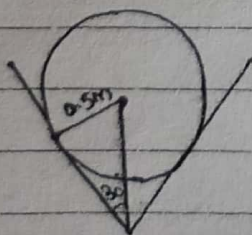
$$R_E \sin 30^\circ + R_F \sin 30^\circ = 500$$

$$\frac{2R_E \times 1}{2} = 500$$

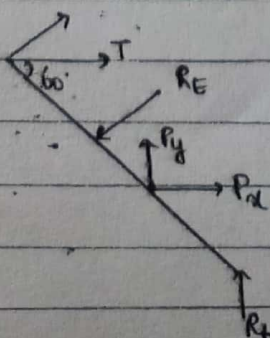
$$R_E = R_F = 500 \text{ N}$$

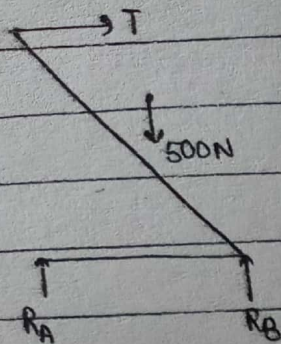


$$\tan 30^\circ = \frac{0.5}{B}$$



$$PE = B = 0.5\sqrt{3}$$





$$R_A = R_B \quad (\text{due to symmetry})$$

$$R_A + R_B = 500$$

$$R_B = 250 \text{ N}$$

Taking moment at P,

$$T \sin 60^\circ \times 1.8 = 500 \times 0.5 \sqrt{3} + 250 \times 1.2 \sin 30^\circ$$

$$T = \frac{432.5 + 300/2}{1.56}$$

~~$$T = 468.5 \text{ N}$$~~

$$T = 373.4 \text{ N}$$